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RESEARCH ARTICLE

OMR-Exam: An Efficient Lightweight Framework for Automated Exam Scoring via YOLO26, EfficientNet, and DCGAN-Enhanced Recognition

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ABSTRACT This study proposes OMR-Exam, a fully automated end-to-end system framework for multiple-choice exam grading on mobile devices. It addresses two challenges in optical mark recognition: reliance on manual coordinate configuration and classification errors caused by data imbalance. We adopted the YOLO26 architecture in conjunction with a Box-in-Box noise filtering algorithm and an Adaptive Grid Sorting technique, enabling the elimination of per-bubble template setup during the coordinate extraction process. In addition, a Deep Convolutional Generative Adversarial Network is employed to synthesize 1,000 artificial samples for the minority crossed-out class. While this generative augmentation demonstrably sharpens the minority-class decision boundary in-domain, cross-template transfer is identified as an open problem. Experimental results evaluated under a 5-fold cross-validation protocol on 735 original answer sheets demonstrate that the core classifier achieves an F1-Score of 89.79% $\pm$ 2.77%. When evaluated at the sheet level in a practical end-to-end pipeline without per-bubble template setup, the proposed framework achieves a mean Sheet Accuracy of 88.98% $\pm$ 3.92%. The statistical analysis demonstrates that while the DCGAN integration yields consistent improvements at the ROI-level for minority classes, the end-to-end sheet-level accuracy gain remains within run-to-run variance, providing a flexible and reliable OMR framework for the automated grading of known exam template formats.

INDEX TERMS Deep convolutional GAN, efficientnet-B0, exam, optical mark recognition, YOLO26.

I. INTRODUCTION

Assessment using Multiple-Choice Questions (MCQs) is one of the common and reliable methods for objectively evaluating students' abilities [1]. However, manual grading is time-consuming, labor-intensive, and prone to human errors [2], [3]. While Computer-Based Testing (CBT) systems provide instant grading capabilities, their large-scale deployment still faces challenges in terms of infrastructure costs, requirements for standardized devices, and cybersecurity risks.

Therefore, paper-based examinations combined with Optical Mark Recognition (OMR) technology remain an effective solution for maintaining fairness and reducing resource consumption. Historically, early OMR systems relied on expensive dedicated scanning hardware, required specialized printed forms, and imposed strict marking rules on users. Later approaches utilized software based on conventional scanners [4]. To overcome cost barriers, modern OMR systems have shifted toward image processing techniques using cameras on mobile devices.

Although flexibility has improved, most OMR systems deployed on mobile devices or in real-world environments still rely on traditional image processing techniques, such as template matching, projection profile analysis, pixel counting, and thresholding. These approaches require fixed templates and are highly sensitive to geometric distortions (e.g., skewed capture angles), poor lighting conditions, and image noise. Moreover, they have a major limitation in that all filled marks are treated as final answers, making it difficult for students to cross out incorrect answers or change their

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selections. To address this issue, the study by Afifi and Hussain [5] redefines OMR from a pixel-counting problem to an image classification problem using Machine Learning (ML) and Convolutional Neural Networks (CNNs), enabling the system to distinguish three states: confirmed, empty, and crossed-out.

Although achieving high accuracy in laboratory environments, the CNN-based approach of M. Afifi and K. F. Hussain [5] and similar methods exhibit two limitations when deployed in real-world scenarios.

First, there is a lack of full automation in the localization stage (Region of Interest (ROI) extraction). The baseline system remains semi-automated, requiring users to manually define the coordinates of answer regions on a reference image. This approach, which relies on feature-based image matching and alignment using Speeded-Up Robust Features (SURF), lacks flexibility when the exam layout changes and is prone to geometric errors when images captured by mobile devices are skewed or affected by perspective distortion.

Second, there is the complexity of the classification architecture and limitations of data augmentation. Large CNN architectures (such as AlexNet) require substantial computational resources. Moreover, these models are prone to overfitting the scanned data. Specifically, when evaluated on images captured using smartphone cameras, the accuracy of traditional CNN models decreased to approximately 60.2%. To mitigate this performance degradation on mobile-captured images and to handle the minority crossed-out class, the baseline study employs a two-stage hybrid architecture combining Bag of Visual Words (BoVW) and CNN. Although this approach improves accuracy to 90.9%, maintaining two independent feature extraction pipelines significantly increases system complexity, making it difficult to optimize inference speed on edge devices. In addition, basic geometric data augmentation techniques are insufficient to simulate the diversity of real human erasure behaviors, necessitating a more advanced synthetic data generation mechanism.

Therefore, this paper proposes OMR-Exam, an efficient lightweight framework for automated exam grading, specifically designed to operate effectively on resource-constrained devices. Instead of relying on manual image alignment, the system utilizes YOLO26, an advanced object detection model, to automatically localize answer regions accurately under various capture angles and paper formats. The extracted regions are then fed into EfficientNet, a highly optimized CNN architecture, to classify answer states with fast inference speed without requiring high-performance hardware. To address the data scarcity of the crossed-out class and to improve the robustness of the model against noise from smartphone cameras, this paper proposes the use of Deep Convolutional Generative Adversarial Networks (DCGAN). By generating diverse synthetic samples, DCGAN helps balance the training dataset and enhances the generalization capability of the system.

The main contributions of this paper are as follows:

1) **Elimination of per-bubble manual configuration:** The proposed system eliminates manual coordinate configuration by integrating the YOLO26 detection network. To fortify this pipeline, we integrated specific adapted implementation choices—namely, a Container Suppression algorithm for removing noisy bounding boxes and an Adaptive Grid Sorting algorithm, enabling the system to automatically interpret the structure of trained exam templates in real time.
2) **Addressing data imbalance using DCGAN:** DCGAN is applied to the OMR task to generate 1,000 synthetic samples for the minority crossed-out class. The generated data simulates diverse human erasure patterns and acts as a strong regularization mechanism, helping to expand the decision boundary of the model.
3) **Improving accuracy using the lightweight EfficientNet-B0 model:** This paper proposes and demonstrates the effectiveness of EfficientNet-B0 as the core classification model. Through experimental validation, the proposed system achieves a mean Sheet Accuracy of 88.98% $\pm$ 3.92%, demonstrating that while the generative augmentation improves ROI-level precision, the end-to-end sheet accuracy differences remain within run-to-run variance, while maintaining efficient inference speed suitable for mobile devices.

The remainder of this article is organized as follows. Section II reviews related studies on Optical Mark Recognition. Section III describes the proposed OMR-Exam method in detail, including the overall system architecture, input representation and preprocessing, automatic localization using YOLO26 and Oriented FAST and Rotated BRIEF (ORB), and answer state classification using EfficientNet-B0. Section IV presents the data synthesis strategy using DCGAN and the model training procedure. Section V describes the experimental setup and provides a detailed analysis of the system performance under various evaluation scenarios. Section VI evaluates and discusses the practical implications, automation capability, contributions of adversarial data generation, and the trade-off between speed and reliability, as well as the current limitations of the system. Finally, Section VII summarizes the findings of this study and outlines directions for future work.

II. RELATED WORK

The development of OMR systems can be categorized into three main technological generations: traditional image processing methods, classical computer vision approaches, and deep learning-based models.

A. TRADITIONAL IMAGE PROCESSING AND CLASSICAL COMPUTER VISION

In early generations, OMR software relied heavily on basic image processing operations such as thresholding, projection profile analysis, pixel counting, and template

matching. Leveraging the characteristics of binarized images, Rakesh et al. [6] developed an OMR system that operates by binarizing answer regions and then applying adaptive thresholding to compute grayscale pixel values for detecting filled answer bubbles. This approach reduces cost by replacing expensive and bulky hardware, while also achieving fast processing speed, capable of grading more than 400 forms in under one minute. However, the system encounters difficulties when marks are erased or when pencil strokes are faint or only partially filled.

In addition, Sattayakawee et al. [7] proposed an automatic grading algorithm for grid-based answer sheets using projection profile techniques and thresholding. The algorithm was evaluated on 300 test samples with a total of 16,500 questions and achieved an average accuracy of 99.909%. However, incorrect detection may occur when ''X'' marks are uneven, erasures are incomplete, or the markings are too faint.

To develop a system capable of operating accurately under challenging real-world conditions, S. Loke et al. [8] employed template matching and morphological dilatation to remove labels inside answer regions. Subsequently, Gaussian adaptive thresholding was applied to detect faint marks, followed by noise reduction using a convolution filter and Random Sample Consensus (RANSAC) to extract linear strokes from noisy points. The system achieved an overall accuracy of 99.94% with a processing speed of 300 cells per second.

In [9], the authors also applied template matching; however, the error rate reached 57.2% for rotated marks. By generating multiple templates with different orientations and scales, or by using rotation-invariant matching, the error rate was reduced to nearly zero. However, this significantly increases computational cost. This technique is most effective when the shape and position of marks are fixed, but it remains sensitive to variations in mark shapes.

Although these methods offer very fast computation and perform well under ideal conditions, they exhibit significant limitations in real-world deployment due to sensitivity to scanning angle deviations, noise, shadows, and unclear markings.

To address these limitations, classical computer vision techniques have incorporated edge detection, contour analysis, Hough Transform, and geometric correction methods (particularly homography) to handle perspective distortion and image skew. Jain et al. [10] used the Canny edge detector on test forms and applied the Suzuki contour tracing algorithm to extract contours of rectangular answer regions. This method is effective even under misaligned input conditions and achieved an accuracy of 99.33% on 500 scanned sheets with a short processing time of approximately 0.48 seconds per sheet.

Hafeez et al. [11] developed a system using the Hough Transform algorithm to automatically detect answer regions and correct image skew. The system then divides the answer sheet into smaller regions and applies pixel projection to determine answers. Although it addresses issues such as skewed or shifted papers, damaged or blurred corner anchors, folded or wrinkled sheets, and faint printing, a software error rate of approximately 0.44% remains and requires further refinement.

One year later, Hafeez et al. [12] extended their work to develop a mobile-compatible version called OMR_FTSC. Instead of using transform-based or individual projection methods, they combined Gaussian blurring with refined preprocessing steps. As a result, images captured by smartphones achieved a significantly improved error rate of 3.33% compared to their previous approach.

In addition, Largo et al. [13] applied gradient-based detection and homography transformation to align answer sheet images on mobile devices. By performing geometric normalization based on anchor points or borders, the system allows recognition algorithms to operate consistently, like data obtained from dedicated scanners, regardless of capture angle. However, these methods typically require complex handcrafted feature design, lack flexibility when the layout of answer sheets changes, and struggle to generalize across different marking styles.

B. DEEP LEARNING AND OBJECT DETECTION IN OMR

Deep learning models have opened a new era for Optical Mark Recognition by automating the feature extraction process and providing improved accuracy and flexibility. Instead of relying on fixed coordinates, many studies have reformulated OMR as an object detection problem using YOLO-based architectures to localize answer bubbles or entire rows of questions, even when the answer sheets are distorted. Tinh and Minh [14] were among the first to implement a YOLO-based system for multiple-choice exam grading, achieving a processing speed of up to 50 sheets per minute with a very low error rate. The study by Tabassum and Rahman [15] further advanced this approach by using YOLOv8n combined with the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm to address the overlap between answer rows, achieving an accuracy of 96.5%. Although these deep learning models can adapt to perspective distortion through data augmentation, in practice they still require a large amount of manually labeled data and often incorporate additional rules based on geometric structure to ensure stability.

In addition, Mahmud et al. [16] proposed a system that uses YOLOv8 to detect and localize two classes on answer sheets, namely ''fill'' (filled or circled answers) and ''cross'' (crossed-out answers) and applies Tesseract Optical Character Recognition (OCR) to extract question indices and read textual answer content. The system achieves high performance with an F1-score of 0.98 and a mean Average Precision (mAP) of 0.99, with an average processing time of approximately 3.5 seconds per sheet. However, the system still encounters difficulties and may produce incorrect

predictions when the input images are of low quality, such as being skewed, blurred, or containing smudged text.

Yookwan et al. [17] combined YOLOv8 with the Cross-Linear Area (CLA) detection technique and a homography algorithm to correct geometric distortion. Experiments conducted on a dataset of 500 real-world answer sheet images, including five different templates captured using smartphones under varying lighting conditions, capturing angles, and distortions, achieved an accuracy of 98.4%.

In [18], the authors trained and compared the performance of three YOLOv8 variants: YOLOv8 Nano, YOLOv8 Small, and YOLOv8 Small Medium. The system handles four common marking scenarios on order forms that reflect real user behavior: blank (no order), crossed (one order), marked with ''X'' (two orders), and crossed-out (canceled order), achieving a classification accuracy of 99.3%. However, the system still has limitations in extraction due to its strong dependence on detecting reference lines.

Thuan et al. [19] used YOLOv8 to detect handwritten ''X'' marks in agree/disagree columns on ballot forms. The data were preprocessed (including sharpening, noise reduction, and contrast enhancement) before being used for detection and recognition. Experimental results achieved an accuracy of up to 98.25% in detecting ''X'' marks, outperforming the use of the TrOCR model alone by 12.5%.

Recently, changes in exam formats, such as the introduction of short-answer questions in the Vietnamese National High School Graduation Examination 2025, have highlighted the limitations of traditional OMR systems. According to [20], the traditional bubble-filling format increases the risk of mechanical errors (e.g., incorrect marking or incomplete erasure), which do not accurately reflect students abilities. To address this issue, the authors proposed a system that allows students to write answers directly and uses the Mathpix Optical Character Recognition (OCR) platform to recognize handwritten text with an accuracy of 98.5%. However, this solution is constrained by the requirement to use a paid commercial Application Programming Interface (API) and the need for input images to be scanned using specialized devices with a minimum resolution of 300 dpi, which poses a significant barrier to deployment on mobile devices.

Shi et al. [21] proposed a Multi-layer Feature Energy Model based on Bayesian Global Optimization to automatically localize, group, and recognize optical mark regions on flexible exam sheets. The dataset consists of 31,940 samples of optical marks with various shapes and arrangements, generated from 110 sets of marking regions distributed to volunteers. The system achieves strong performance, with Single-mark Localization and Grouping reaching 97.07% and 97.86% precision, respectively, and a Recognition Accuracy of 97.80%. The results indicate that the model is not constrained by rigid exam templates or fixed mark shapes and demonstrates robustness against distortions on paper. However, the Bayesian optimization process requires substantial computational resources and processing time.

Hasan et al. [22] proposed the use of the Modified Bidirectional Associative Memory (MBAM) technique for real-time OMR recognition. The system can detect cases where candidates leave answers blank or select multiple answers simultaneously without the need to segment or extract individual answer regions. The system achieves an average accuracy of up to 99.988%. When tested on 1,000 answer sheets containing a total of 72,500 questions, the system correctly recognized 72,492 questions and failed to recognize only 8. The average processing time per sheet is relatively fast, taking less than 30 seconds. However, the system requires converting the entire image matrix of the answer sheet into a bipolar representation, where each pixel is assigned a value of either 1 or -1 before being processed by the MBAM network. In addition, the system still relies on an Automatic Document Feeder (ADF) scanner to acquire JPG images, rather than being fully optimized for capture using smartphone cameras, which are currently a more flexible trend.

C. LIGHTWEIGHT CLASSIFICATION AND THE CHALLENGE OF DATA SCARCITY

After the localization stage, CNNs are commonly used to classify the state of each answer bubble (e.g., empty, filled, or crossed-out) [23]. To meet the demands of large-scale processing and deployment of resource-constrained devices (such as mobile devices), the use of lightweight deep models is essential. Previous studies have shown that models such as MobileNetV2 can maintain high accuracy while ensuring real-time inference performance [24].

Despite achieving promising results, current deep learning-based solutions still face significant challenges related to data scarcity due to the lack of publicly available, diverse, and large-scale OMR datasets [25]. This leads to overfitting, causing a noticeable drop in model accuracy when encountering datasets with different capture conditions or printing quality. To address this bottleneck, synthetic data generation for augmenting training datasets has been identified as an essential research direction to improve the generalization capability of OMR models.

To address the major barrier of deep learning, which is the requirement for extensive data collection and annotation, Hernández-Mier et al. [26] proposed a fully unsupervised OMR algorithm, specifically designed to process and extract answers from large-scale printed multiple-choice exams. The system achieved an overall accuracy of up to 99.95% when evaluated on 564,040 answers, with only 424 misclassified cases. The processing speed is also notable, with an average of 1.05 seconds per exam (equivalent to 57 exams per minute). The total time required to process 6,029 exams was 1 hour and 45 minutes. However, the system still shows performance degradation (although remaining above 80%) when handling cases where multiple answers are marked simultaneously, and it encounters difficulties with physical defects such as severely misoriented scanned images, folded

paper, shadow noise from scanners, or unintended markings overlapping the answer regions.

Similarly, in the context of unsupervised learning, Shi et al. [27] combined their approach with an energy optimization method to automatically localize optical marks. The Single Mark Localization Model (SMLM) is formulated and optimized based on three main energy components: distributed energy, proportional (duty) energy, and aspect ratio energy. The study evaluated single-mark localization and recognition on a test set of 268 answer sheets, each containing 60 optical marks. For 242 sheets (accounting for 90% of the total), both precision and recall reached 100%. For the remaining 10%, precision and recall were maintained above 95%.

In addition, Sancar et al. [28] employed an unsupervised machine learning approach based on the K-means clustering algorithm, focusing on analyzing individual marking density to improve grading speed and accuracy. Instead of using a fixed threshold, the algorithm automatically adapts to each candidates marking style and pressure (i.e., intensity of marks) to accurately distinguish between filled and empty bubbles. The real-world dataset consists of 265,816 multiple-choice answer sheets. The system achieved an accuracy of 99.98%, with a real-time processing speed of 0.12 seconds per sheet. Although the system demonstrates stable performance at large scale, the initial preprocessing stage still requires manual operations to create configuration files, and the extraction process may fail if the paper is folded, torn, or if the QR code is smudged.

Palak et al. [29] proposed an image processing algorithm implemented in Python using the OpenCV library. Instead of using complex deep learning models, this method extracts information by storing the indices of filled bubbles as values and the question numbers as keys in a dictionary structure. The system then compares the students answer dictionary with the ground-truth answer dictionary to compute scores. The system achieved an accuracy of 95.67% under real-world conditions, with a processing time of 4 seconds per answer sheet. This algorithm is designed to be scalable and capable of handling any number of questions. However, it still encounters limitations when images are captured with excessive skew, when markings are unclear, when ink spills outside the bubbles, or when erasures are incomplete.

The OMR-Exam framework proposed in this study is developed to address these research gaps by combining the speed and strong localization capability of YOLO26, efficient extraction and classification of answer states using the lightweight EfficientNet architecture and mitigating the scarcity of real-world data through DCGAN to enhance the generalization capability of the system.

III. PROPOSED METHOD

A. OVERALL ARCHITECTURE

To ensure high automation, accuracy, and real-time processing capabilities, the system is designed to operate as a closed-loop sequential pipeline. A defining characteristic of our proposed architecture is the decoupling of spatial localization from per-sheet processing. Rather than running the object detection network on every incoming exam sheet, YOLO26 is executed only once on a clean, blank template for each exam format to generate a Master Coordinate Map. For subsequent student sheets, the system bypasses heavy neural network detection entirely, relying instead on the lightweight ORB algorithm to geometrically register and project the sheet onto the Master Coordinate Map. The data flow progresses through five processing stages, as illustrated in Figure 1.

1) STAGE 1: TEMPLATE INITIALIZATION & LEARNING

This is the dynamic configuration step, enabling the system to adapt to provided exam structure formats (from a single page to multiple pages with varying layouts) without manual per-bubble intervention. This stage is conducted in three steps as follows:

1) First, the exam configuration is required, where the user actively provides a standard template image and declares parameters such as the number of pages, questions per page, choices per question (e.g., 4 options A, B, C, D), and the Answer Key.
2) Subsequently, automated localization is performed using the YOLO26 network, which scans the entire template image to extract the spatial coordinates of the multiple-choice boxes. To mitigate the false detection of question block borders (Ghost Boxes), the Container Suppression (Box-in-Box noise filtering) algorithm is employed to eliminate overlapping bounding boxes.
3) Finally, Grid Mapping is executed, where the collected raw coordinates are passed through the Adaptive Grid Sorting algorithm. This algorithm automatically aligns and sorts the boxes in geometric order (top-to-bottom, left-to-right) while assigning logical identifiers (e.g., Q1_A, Q1_B) to construct an accurate Master Layout Map.

2) STAGE 2: SCANNING & GEOMETRIC ALIGNMENT

The system receives a batch of student answer sheet images from mobile devices. These images are required to be stored in a static naming convention: *[StudentID]_[ExamCode]_[Page].extension* (e.g., *10299_exam0_1.jpg*) to facilitate the automated processing pipeline. The data is then analyzed; the system automatically extracts the filename to identify the student (StudentID), exam code, and corresponding page number to invoke the correct Layout Map.

Actual answer sheet images captured by mobile devices often contain visual noise and perspective distortion due to tilted shooting angles, varying distances, or rotated paper. Therefore, we utilize the ORB feature extraction algorithm; both the answer sheet image and the template image are converted to grayscale space. This algorithm extracts thousands of invariant keypoints and represents

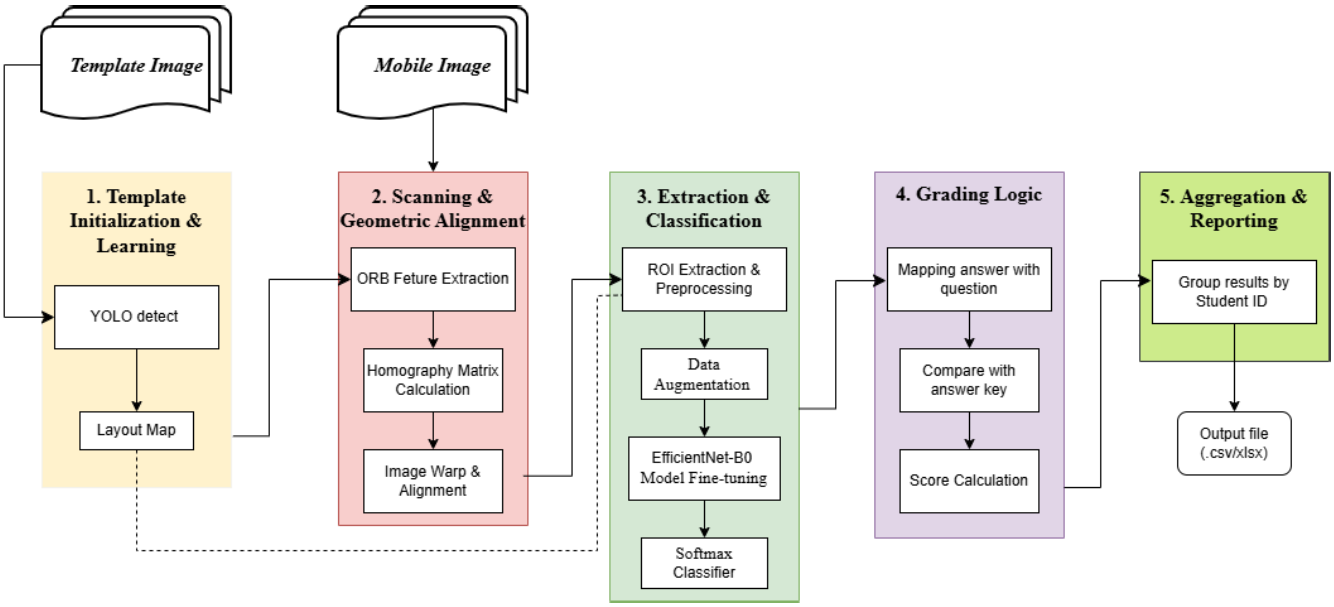


FIGURE 1. Overall workflow of the proposed system.

them as binary descriptors. Next, spatial synchronization is performed using a Brute-Force feature matcher with Hamming distance; the system identifies corresponding point pairs between the two images. From this set of points, a Homography transformation matrix (H) is calculated. The Warp Perspective transformation is applied to flatten and align the answer sheet image, making it perfectly coincident with the spatial coordinate system of the Master Layout Map.

3) STAGE 3: EXTRACTION & CLASSIFICATION

This is a critical stage in converting raw optical data into logical state information, leveraging the power of specially optimized Deep Learning architectures. The process is carried out through the following steps:

1) **ROI Extraction & Preprocessing:** The Layout Map generated in Stage 1 is directly applied to the aligned answer sheet image plane from Stage 2. The system automatically crops and extracts a batch of Regions of Interest (ROIs) containing independent multiple-choice boxes based on precise bounding box coordinates. After cropping, the small ROI patches are resampled using Bilinear Interpolation to an input size of 128×128 pixels.
2) **Data Augmentation & Model Fine-tuning:** To ensure the classification model achieves maximum accuracy during the inference phase, a rigorous two-stage training strategy is applied to overcome the extreme data imbalance of the *crossed-out* class. Specifically, starting with approximately 120 original real-world crossed-out samples per training fold, the system first applies automated geometric and morphological transformations (including random rotation, random zoom, local distortion, and shear mapping) to expand the foundational sample space to exactly 1,500 images. Subsequently, DCGAN is trained on this dataset to synthesize an additional 1,000 morphologically diverse artificial samples. The core EfficientNet-B0 learning network is then fine-tuned on this expanded hybrid dataset (totaling 2,500 samples for the minority class per fold), successfully fortifying the decision boundary and mitigating model bias.
3) **Softmax Classification:** In the actual execution pipeline, the batch of ROI images (128×128 pixels) is passed through the fine-tuned EfficientNet-B0 network. The neural network analyzes the features and outputs a probability distribution vector via the Softmax function. Based on the highest probability value, the system finalizes the state identity of each box into one of three labels: Confirmed, Crossed-out, or Empty, serving as the prerequisite for the grading logic in Stage 4.

4) STAGE 4: GRADING LOGIC

The discrete state data of the multiple-choice boxes is mapped back to the logical structure of the exam to extract the candidate's final selection. The boxes are grouped by question (e.g., the group of 4 boxes Q1_A, Q1_B, Q1_C, Q1_D constitutes Question 1). The Rule Engine initiates processing:

- Crossed-out states are discarded (considered as the candidate canceling that answer).
- If there is only one Confirmed box, the system records it as the candidate's choice.
- If there is more than one Confirmed box (double shading) or no box is confirmed (Empty), the question is marked as invalid/incorrect.

After processing, the candidate's valid choices are directly compared to the Answer Key. The system accumulates the score based on a pre-configured grading scale to produce the raw score for the current page.

5) STAGE 5: AGGREGATION & REPORTING

This stage handles data integrity at a large system scale, particularly for multi-page exams. Since an exam may consist of multiple pages, the detailed scores and states of each individual page image are temporarily stored in the cache, strictly indexed by the Student ID.

Once the scanning pipeline concludes, the system automatically performs a Group-by operation based on the Student ID. An integrity-checking algorithm ensures that the candidate has submitted the complete set of constituent pages (no missing or extra pages).

B. INPUT REPRESENTATION AND PREPROCESSING

Image data acquired from mobile device cameras is often heavily impacted by uncontrolled environments, such as uneven lighting, localized shadows, and sensor noise. To ensure stability for the deep learning models, the input representation process is standardized through three specialized processing steps.

1) DETECTION INPUT STREAM

The system's first step involves loading the Template image into the YOLO26 network to extract the coordinate map of the multiple-choice boxes. The input to the YOLO26 network is normalized to a size of 960 $\times$ 960 pixels. To preserve the geometric features of the circular multiple-choice boxes and prevent aspect ratio distortion, a padded resizing (letterboxing) mechanism is applied, where the redundant areas are filled with a neutral color value. This technique guarantees that regardless of the input image's aspect ratio (e.g., A4 or Letter), the physical shapes of the target objects remain intact.

2) COLOR SPACE CONVERSION AND NOISE REDUCTION

When the system receives the candidate's actual answer sheet images, they are frequently skewed or rotated due to the capture operation. To align the answer sheet image with the standard template, the system utilizes the ORB algorithm. At this computational milestone, the image is temporarily compressed from 3 color channels to a single Grayscale channel to reduce the matrix computation load by two-thirds and to provide resistance against lighting noise. The conversion formula employs the ITU-R BT.601 standard luminance weights [30], as described in Equation (1):

$$Y = 0.299R + 0.587G + 0.114B \tag{1}$$

where Y is the pixel intensity value at position (x, y).

After successfully extracting features and calculating the Homography matrix on the grayscale space, the system applies this matrix to the original color image of the answer sheet. Consequently, the output of this stage is an aligned answer sheet image that retains its 3 color channels intact to serve the subsequent deep classification phase.

3) CLASSIFICATION INPUT STREAM

The system utilizes the coordinate map extracted from stage 1 (template initialization and learning), to directly crop the Regions of Interest (ROIs) from the aligned color answer sheet image. The cropped ROIs containing the multiple-choice boxes have non-uniform sizes due to varying camera distances. Before being fed into the core classification network (EfficientNet-B0), these image patches are processed as follows:

- **Resampling:** Bilinear Interpolation is used to standardize all cropped ROIs to a resolution of 128 $\times$ 128 pixels (instead of the standard ImageNet 224 $\times$ 224). As an empirical trade-off observed during our experiments, this specific size balances inference speed and feature retention. It acts as a gentle smoothing mechanism that effectively preserves the structural textures of the pencil marks without introducing the destructive blur of over-interpolation, while simultaneously minimizing GPU memory bandwidth.
- **Z-score Normalization:** To leverage pre-trained weights on the ImageNet dataset, the pixel intensity $I(x, y)$ is normalized according to the standard normal distribution as shown in Equation (2) [31]:

$$I_{norm}(x, y) = \frac{I(x, y) - \mu}{\sigma} \tag{2}$$

where the mean vector is $\mu = [0.485, 0.456, 0.406]$ and the standard deviation is $\sigma = [0.229, 0.224, 0.225]$ corresponding to the three RGB color channels. This step synchronizes the input data distribution, allowing the model to converge faster and extract more accurate features right from the initial layers.

C. AUTOMATED LAYOUT ANALYSIS VIA YOLO26 AND ORB

This module is responsible for entirely replacing the manual coordinate entry operations found in previous studies.

1) AUTOMATED ROI DETECTION

We deploy YOLO26—the newly released version of the YOLO object detection network architecture—as the core spatial localization engine for the system, specifically utilizing the yolo26s model to optimize the balance between speed and performance. The breakthrough of YOLO26 lies in its NMS-Free architecture and its Small-Target-Aware mechanism [32]. In the context of the OMR problem, where numerous tiny multiple-choice boxes are densely distributed across a large paper surface, the Small-Target-Aware Label Assignment (STAL) mechanism allows the model to accurately extract features without degradation through deep convolutional layers. Concurrently, the complete elimination of the Non-Maximum Suppression (NMS) process helps the system thoroughly optimize inference latency, excellently

meeting real-time processing requirements on edge AI devices like smartphones.

Although YOLO26 demonstrates sensitivity and stability on most of the dataset, empirical testing reveals that this architecture is not entirely immune to a minority of edge cases characterized by the specific geometric traits of exam papers. The model may misclassify the borders encompassing an entire question block (*Container/Block borders*) as a large boundary enclosing multiple valid answers inside. Standard IoU-based filtering struggles with this hierarchical overlap (*Box-in-Box*) because the smaller, valid boxes are often completely swallowed by the larger container.

Algorithm 1 Container Suppression

Require: A set of bounding boxes $B = \{b_1, b_2, \ldots, b_n\}$, where each bounding box $b_i = (x_i^{\min}, y_i^{\min}, x_i^{\max}, y_i^{\max})$

Ensure: A refined set of bounding boxes D, obtained by removing noisy or irrelevant bounding boxes from B.

$R \leftarrow \emptyset$ // Initialize the set of indices corresponding to bounding boxes to be removed
$N \leftarrow |B|$ // Assign N as the number of elements in B
for $i \leftarrow 1$ to N **do**
  for $j \leftarrow 1$ to N **do**
    if $i = j$ **then continue**
    if $b_j \subset b_i$ **then**
      $R \leftarrow R \cup \{i\}$ // Add index i to the set R
      break
    end if
  end for
end for
$D \leftarrow \emptyset$ // Initialize the output set of bounding boxes
for $k \leftarrow 1$ to N **do**
  if $k \notin R$ **then**
    $D \leftarrow D \cup \{b_k\}$ // Add bounding box b_k to the set D
  end if
end for
return D

1) **Dynamic Confidence Sweeping:** Instead of relying on a rigid baseline, we implemented a dynamic confidence thresholding strategy. The system iteratively sweeps the confidence level downwards (e.g., 0.50, 0.40, 0.30, 0.25, 0.15). The loop evaluates detections and automatically terminates the search once the exact number of required template bubbles is identified, guaranteeing high-fidelity extraction while minimizing false positives.
2) **Median-Area Heuristic Filtering:** Before complex geometric checks, a preliminary size filter is applied. By calculating the median area of all raw detected boxes, bounding boxes whose areas fall outside the heuristically defined valid range ([0.4 × median, 2.5 × median]) are immediately discarded. This step effectively eliminates extreme outliers such as tiny ink specks or massive page borders.
3) **Container Suppression (Algorithm 1):** To resolve the "Box-in-Box" artifacts, a strict geometric containment check is applied with a padding tolerance of $\epsilon = 5$ pixels. For any two distinct bounding boxes $b_i, b_j \in B$, if the coordinates of b_j are geometrically contained within the spatial boundaries of b_i, the significantly larger box b_i (the false container) is systematically discarded. Because valid answer boxes are uniformly small, there are physically no valid large enclosing boxes on the sheet, ensuring a reverse false-removal rate of exactly 0%.
4) **Area-Variance Fallback:** In rare instances where the number of surviving boxes still exceeds the target quantity after the aforementioned filters, a mathematical fallback is triggered. The system resolves the surplus by calculating the absolute difference between each box's area and the median area. The exact number of required boxes exhibiting the minimum variance is retained, strictly isolating the most morphologically consistent multiple-choice bubbles.

This multi-stage containment and filtering rule acts as a strict geometric gatekeeper, ensuring the system only retains coordinates converging closest to the actual physical margins of each answer bubble, effectively eliminating the need for manual coordinate configuration.

2) GRID STRUCTURE ANALYSIS & GEOMETRIC ALIGNMENT

This process serves as a bridge between object detection and data extraction, executed sequentially in two steps: establishing a coordinate map on the template image and synchronizing space on the answer sheet image.

Step 1: Adaptive Grid Sorting on Template

Immediately after YOLO26 extracts the set of Bounding boxes from the standard template image, these coordinates are initially in an unstructured state. To establish a Master Coordinate Map, they must be sorted in a logical order (Row-by-Row). However, mechanical printer errors or YOLO's quantitative positioning variances often prevent boxes on the same line from being perfectly aligned (Y-axis deviation error). Instead of using a fixed pixel distance, we employ the Adaptive Sorting Threshold algorithm (θ_y). This value scales automatically based on the size statistics of all detected boxes, as described in (3):

$$\theta_y = \lambda \times \frac{1}{|B|} \sum_{i=1}^{|B|} (y_{max}^i - y_{min}^i) \tag{3}$$

where λ is an empirical coefficient. While this general row-grouping heuristic aligns with standard spatial methodologies in document layout analysis [7], [21], the specific optimal value of $\lambda = 0.6$ was established empirically through our ablation study. As demonstrated by the sensitivity sweep from 0.4 to 0.8, this value serves as a conservative safe margin—large enough to neutralize intra-row Y-axis deviations, but strictly constrained to prevent the risk of merging adjacent rows in densely packed templates. Any two boxes b_i, b_j

are assigned to the same logical row if the vertical distance between their centers is less than this threshold: $|y^i_{center} - y^j_{center}| < \theta_y$. The output is a labeled identifier matrix (e.g., Q1_A, Q1_B, ...).

Step 2: ORB Geometric Alignment

The actual student's answer sheet image ($I_{student}$) is often subjected to perspective distortion (skewed, rotated) compared to the template image ($I_{template}$). For synchronization, we utilize the ORB algorithm [33].

- Feature Extraction: The system searches for invariant Keypoints on both images.
- Matching: The binary Descriptors are matched via Hamming distance to identify the corresponding point pairs $P_{matches}$.

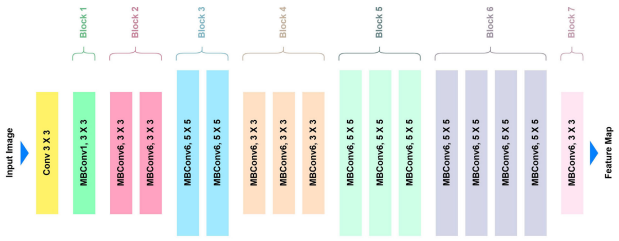


FIGURE 2. EfficientNet-B0 architecture [31].

- Transformation: A homography matrix H (3×3) is calculated based on $P_{matches}$. To accurately align the images, perspective transformation is applied via an inverse warping process. For each target pixel coordinate (x_d, y_d) in the aligned output image $I'_{student}$, its corresponding homogeneous coordinate in the source image $I_{student}$ is calculated using the inverse homography matrix H^{-1}:

$$\begin{bmatrix} x_s \\ y_s \\ w \end{bmatrix} = H^{-1} \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} \tag{4}$$

The final pixel value is then determined through bilinear interpolation at the normalized coordinates $(x_s/w, y_s/w)$:

$$I'_{student}(x_d, y_d) = \text{Interpolate}\left(I_{student}, \frac{x_s}{w}, \frac{y_s}{w}\right) \tag{5}$$

This process aligns the answer sheet image to coincide perfectly with the template's coordinate system. Finally, the Master Coordinate Map (from Step 1) is directly applied to the aligned answer sheet image ($I'_{student}$) to accurately crop the ROIs containing the answer boxes, preparing them for the classification step.

D. CLASSIFICATION USING EFFICIENTNET-B0

Following data preprocessing, the inputs are fed into the classification model. The system necessitates an architecture capable of extracting fine-grained features (such as thin ink strokes, faint pencil marks, or residual eraser smudges) while strictly maintaining a high inference speed to process thousands of ROI patches in real-time. To meet these stringent criteria, we propose utilizing EfficientNet-B0 as the core learning model, as illustrated in Figure 2.

Designed through Neural Architecture Search (NAS) [31], EfficientNet-B0 is a baseline model that achieves an optimal balance between accuracy and computational cost (FLOPs). This neural network possesses a streamlined structure with a total of approximately 5.3 million parameters, sequentially divided into 9 processing stages [31]:

- **Stage 1 (Stem):** Initiates with a standard 3×3 convolutional layer to extract fundamental surface features of the input image.
- **Stages 2 to 8 (Body):** This constitutes the core of the architecture, comprising 16 Mobile Inverted Bottleneck Convolution (MBConv) blocks. Depending on the stage, these MBConv blocks alternately utilize 3×3 and 5×5 kernel filters, combined with a flexible expansion ratio of 1 or 6. The core characteristic of MBConv is the application of Depthwise Separable Convolution, which substantially minimizes matrix computation volume compared to traditional convolution without degrading feature representation quality.
- **Stage 9 (Head):** Concludes with a 1×1 convolutional layer, followed by a Global Average Pooling layer and a Fully Connected Layer to perform the multi-class classification task.

Tightly integrated within these MBConv blocks is the Squeeze-and-Excitation (SE) channel attention mechanism [31]. This mechanism acts as an intelligent signal filter, automatically squeezing global information and exciting crucial channel weights. This process allows the model to automatically enhance the weights for feature maps containing the geometric information of pen strokes, while suppressing the weights of channels carrying noise from the paper background. Consequently, the system significantly improves accuracy when classifying states with easily confused visual features, such as the crossed-out and Confirmed labels.

The breakthrough of the EfficientNet architecture family lies in the Compound Scaling method [31]. This technique does not scale parameters discretely; rather, it uses a fixed scaling coefficient to uniformly scale all three dimensions of the network: depth, width, and input image resolution. Although the EfficientNet family offers variants from B1 to B7 with massive capacities, we intentionally designed our choice around EfficientNet-B0—the lightweight baseline version. Despite its ultra-compact architecture, B0 still excellently meets, and even exceeds, accuracy expectations for the OMR problem based on three primary reasons:

1) **Distinctive Data Feature Representation:** The ROI images of multiple-choice boxes possess a relatively monotonic geometric structure, with visual complexity far lower than real-world contexts in the ImageNet dataset. The lightweight B0 model provides just the right amount of feature extraction power, avoiding

representational redundancy and eliminating the overfitting risk commonly encountered in excessively deep networks (like B4–B7).

2) **Resource Optimization:** EfficientNet-B0 consumes only about 5.3 million parameters (more than 12 times smaller than the 66 million parameters of the heaviest B7 version). This drastically minimizes the required Video RAM (VRAM) capacity, optimizing the training and practical deployment processes.
3) **Deployability on Low-End Devices:** The focal point of the OMR system is its ability to operate smoothly on devices with constrained computational resources. With low inference latency, the lightweight EfficientNet-B0 model overcomes the speed barrier when processing batches of cropped images in real-time, helping the system achieve high accuracy while ensuring feasibility for real-world deployment.

IV. DATA SYNTHESIS AND TRAINING STRATEGY

To construct a robust OMR system, the development process is structured into three distinct phases: (1) Identifying and addressing the data scarcity challenge in OMR, (2) Augmenting data via the GAN model, and (3) Detailing the optimization and training process, which includes the answer box bounding model and the fine-tuning of EfficientNet-B0.

A. THE CHALLENGE OF DATA SCARCITY IN OMR

During the construction of the classification model, a major challenge arises regarding class imbalance. Specifically, the system inherits the real-world multiple-choice question (MCQ) dataset published in the original study by Afifi and Hussain [5]. Statistics from this dataset accurately reflect candidate behavior in actual testing environments: the vast majority of multiple-choice boxes are always in the Empty (unshaded) or Confirmed (validly shaded) state. The original dataset records an overwhelming distribution disparity with 22,367 Empty samples and 10,980 Confirmed samples. Conversely, crossed-out behavior is a rare event, occurring only when candidates make a mistake and need to revise their answer, resulting in a mere 202 original samples being collected.

With this massive discrepancy—over 33,000 samples in the majority classes compared to roughly 200 samples in the minority class—significant difficulties are posed for the neural network optimization process. The original authors, Afifi and Hussain [5], explicitly acknowledged this severe imbalance, as the actual crossed-out class accounts for only about 0.6% of the total samples. To mitigate this impact, the baseline study had to intervene by adjusting the prior probability while simultaneously applying basic geometric transformations (e.g., translation, rotation, flipping) to augment the training data. However, confronting the colossal disparity between the majority and minority classes, these traditional interventions may still fail to thoroughly resolve the difficulties in optimizing deep neural networks.

B. DATA AUGMENTATION VIA GAN MODELS

To overcome the class distribution imbalance, where the crossed-out class constitutes an infinitesimally small minority compared to the other two classes, we propose a two-stage data augmentation pipeline, combining targeted geometric transformations with a DCGAN. To clarify the scope of our generative pipeline, DCGAN was exclusively applied to the minority crossed-out class. The empty and confirmed classes already overwhelmingly dominate the dataset (over 33,000 samples combined). Synthesizing additional samples for these majority classes would merely inflate computational resource consumption without addressing the core bottleneck of class imbalance. While traditional geometric transformations provide baseline variance, comprehensive surveys on image data augmentation [34] and empirical studies in critical domains [35] have demonstrated that generative augmentation is far superior for addressing severe minority class problems. These studies prove that GANs can synthesize fundamentally novel structural features rather than merely perturbing existing pixels, anchoring our methodological choice to prefer DCGAN over simpler synthesis approaches.

Step 1: Non-linear Geometric Augmentation

Under the strict 5-fold cross-validation isolation, the training set of each fold begins with a highly constrained sample space (approximately 120 actual crossed-out images). To prevent overfitting during GAN training, we apply an automated sequence of multi-morphological spatial transformations, explicitly configured to simulate natural physical and optical deviations in test paper scanning. These include random rotation (simulating tilted paper), random zoom (simulating varying stroke thickness), local distortion (simulating paper creases), and shear mapping (simulating perspective skew). This process amplifies the foundational dataset to exactly 1,500 images per fold. This collection serves as the essential seed data, providing a robust structural foundation to initialize the adversarial generative network.

Step 2: DCGAN Architecture & Optimization

To enable the system to generalize and recognize anomalous erasure behaviors, we designed and trained a specialized DCGAN architecture [36] tailored to the prepared 1,500-image seed dataset.

Generator Network (G): The model receives a normally distributed random noise vector $z \in \mathbb{R}^{100}$ (latent space). This vector sequentially passes through 4 Transposed Convolution blocks to decode and amplify the tensor space. To ensure gradient stability, each convolutional layer is coupled with a Batch Normalization layer and a ReLU activation function. At the output layer, the tanh activation function is applied to map the pixel space to the $[-1, 1]$ range, producing a synthetic output image with standard dimensions of 64×64 pixels and 3 color channels.

Discriminator Network (D): Counteracting G, the D Network acts as a rigorous geometric feature extractor, comprising a series of 4 Convolutional layers (Conv2D). The architectural breakthrough here is the use of a stride $s = 2$ for downsampling instead of standard Pooling layers, allowing

the network to preserve the fine spatial geometries of the ink strokes. To thoroughly prevent the vanishing gradient phenomenon, a LeakyReLU activation function with a leak coefficient $\alpha = 0.2$ is applied across the hidden blocks. Additionally, Dropout layers ($p = 0.3$) are injected to prevent discriminator overfitting on the constrained 1,500-sample space. At the output layer, a Sigmoid function estimates the probability to distinguish between real and synthetic images.

Training Strategy: The adversarial training cycle is executed for 300 epochs. Instead of a uniform learning rate, we implemented the Two Time-Scale Update Rule (TTUR). The Generator's learning rate is set to $\eta_G = 0.0002$, while the Discriminator is intentionally restricted to $\eta_D = 0.00005$. Both networks are optimized using the Adam optimizer with a momentum $\beta_1 = 0.5$. This asymmetrical update mechanism successfully prevents the Discriminator from converging too rapidly and overpowering the Generator, maintaining optimal equilibrium in the min-max game.

Step 3: Quantitative Evaluation and Dataset Synthesis

To provide quantitative evidence of the synthetic dataset's diversity, we implemented an automated evaluation pipeline utilizing the Fréchet Inception Distance (FID) across our 5-fold cross-validation setup.

First, to ensure an apples-to-apples measurement space, the 1,500 real crossed-out images in each fold were aligned to the Generator's output specifications (using Square Padding and resizing to 64 × 64) to prevent any resolution-induced discrepancies. Rather than arbitrarily selecting the final epoch (Epoch 300), we designed an automated FID scanning strategy. For every saved checkpoint, the Generator synthesized 1,000 temporary images, and the FID score was computed against the 1,500 training images. The checkpoint with the lowest FID score was dynamically selected as the ''Optimal Model'' for that fold.

However, to rigorously measure the out-of-sample representativeness independently from the training augmentation distribution, we computed the final evaluation strictly against the held-out, unaugmented real *crossed-out* samples in the Testing Set of each fold. The dynamically selected optimal generators yielded the following out-of-sample per-fold FID values: Fold 1 (201.26), Fold 2 (197.87), Fold 3 (208.75), Fold 4 (210.55), and Fold 5 (205.87), resulting in a mean FID of 204.86 ± 5.25.

While an absolute FID of ∼204 might appear relatively high, we analyzed this metric within our specific domain constraints:

- **Sample Size Bias (Small-N):** FID is highly sensitive to sample sizes. Calculating FID on a heavily constrained held-out test space (only 30 to 40 images per fold) inherently inflates the absolute mathematical value.
- **Feature Domain Mismatch:** The Inception-v3 network is pre-trained on ImageNet (colorful, complex natural scenes), which creates a significant domain gap when evaluating simplistic, binary-like optical marks.

Despite this baseline shift, the tightly constrained variance across all five folds (±5.25) quantitatively confirms the stability of the generator's out-of-distribution synthesis.

Furthermore, to strictly rule out the risk of training data memorization, we implemented a Nearest-Neighbor Memorization Check. We measured the Structural Similarity Index (SSIM) between the 1,000 synthetic images and the original, unaugmented real training set (prior to any Stage 1 geometric augmentations). Analyzing the optimal generator in Fold 1, the average SSIM between generated images and their closest real counterparts was exceptionally low at 0.3610, with the absolute maximum similarity peaking at only 0.8385. The complete absence of near-identical pairs (SSIM $\approx$ 1.0) quantitatively proves that the generator successfully synthesized novel morphological erasure features rather than memorizing the limited original training distribution.

Finally, using the dynamically selected Optimal Model for each fold, the Generator synthesized exactly 1,000 high-quality artificial samples. This augmented data volume is subsequently integrated into the initial training set, thereby resolving the class distribution imbalance problem and establishing an optimal prerequisite for fine-tuning the classification architectures.

To bridge the resolution gap between the native DCGAN output (64 × 64 pixels) and the core classifier's input requirement (128 × 128 pixels), Bilinear Interpolation is applied. Upscaling to 128 × 128 provides an empirical trade-off compared to the standard ImageNet 224 × 224 resolution, balancing inference speed and feature retention. As illustrated in Fig. 3, the bilinear filter at 128 × 128 softly smooths the raw blockiness of the 64 × 64 GAN output without causing the destructive washed-out blur seen at 224 × 224. This ensures that the interpolated synthetic images remain visually coherent, preserving the essential geometric semantics of the crossed-out strokes for the CNN to learn effectively.

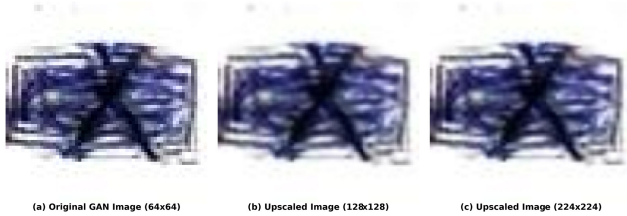


FIGURE 3. Visual coherence comparison of DCGAN-generated crossed-out samples across different upscaling resolutions: (a) Original GAN output (64 × 64), (b) The proposed optimal interpolation (128 × 128), and (c) Standard ImageNet resolution (224 × 224).

C. OPTIMIZATION AND TRAINING DETAILS

1) TRAINING THE ANSWER BOX BOUNDING MODEL

The objective of this phase is to train the neural network to accurately recognize and localize the spatial coordinates (bounding boxes) of numerous small multiple-choice bubbles distributed across the entire exam paper surface.

We inherit the dataset comprising 735 answer sheets totaling 815 pages (including 6 different exam formats from

Exam 0 to Exam 5) as described in the original study by Afifi and Hussain [5]. This dataset encompasses high structural diversity, containing both fully completed and blank sheets, thereby compelling the model to learn the invariant geometric features of the multiple-choice box rims rather than being distracted by the ink marks inside.

The dataset comprising 815 original pages is strictly partitioned at the page level into a 70% Training set, a 20% Validation set, and a 10% Testing set. This ensures the YOLO26 model is evaluated on a fully unseen partition.

Crucially, to prevent any data leakage during the end-to-end grading evaluation (Scenario B), this trained YOLO26 model does not perform inference directly on the actual student answer sheets in the testing sets. Instead, the detection model is executed exactly once on 6 completely blank template sheets (which contain no student marks and are separate from the student dataset) to extract a Master Coordinate Map. Geometric alignment algorithms are then used to map these master coordinates onto the test sheets. This automated template extraction mechanism ensures that the test sets remain completely unseen to the detector during inference.

Unlike previous generations, YOLO26 is optimized through the Small-Target-Aware Label Assignment (STAL) mechanism combined with an Advanced Bounding Box Regression Loss function. This training process enables the neural network to highly optimize its localization accuracy, minimizing pixel coordinate errors and demonstrating strong baseline metrics for mean Average Precision (mAP@50 and mAP@50–95) for extremely small objects like multiple-choice bubbles.

Specifically, when evaluated on the 10% held-out testing set, the YOLO26s model demonstrated exceptional localization capabilities for tiny objects, achieving a Precision of 99.81%, a Recall of 99.82%, a mAP@50 of 99.49%, and a mAP@50-95 of 94.24%.

2) TRAINING CLASSIFICATION MODELS

To evaluate and select the most optimal architecture, we conduct empirical training on advanced image classification models, including EfficientNet-B0, YOLO26s-cls, and ResNet50. The entire training data consists of individual ROI patches extracted directly from the original exam sheet dataset. Rather than approaching this via traditional machine learning methods that are prone to bias due to the natural imbalance of testing data, our training strategy focuses on three core methodologies:

1) **Data Preprocessing and Spatial Normalization:** Because extracted ROIs vary significantly in aspect ratio due to bounding box tolerances, directly distorting them into squares would destroy the morphological features of pencil strokes. Therefore, a uniform preprocessing pipeline is applied. First, *White-Square Padding* (RGB: 255, 255, 255) is applied to all ROIs to preserve the aspect ratio while seamlessly blending the padding with the exam paper background, preventing the convolutional filters from detecting fake edges. Subsequently, *Micro-aware Resizing* is applied. Unlike natural image classification that uses large resolutions (e.g., 224 × 224), OMR bubbles are physically small. Upscaling them excessively inflates computational cost and risks blurring surface textures. Thus, all inputs are normalized to an optimal resolution of 128×128 pixels (`imgsz=128`). For EfficientNet-B0 and ResNet50, the inputs are further standardized using ImageNet mean and standard deviation distributions.
2) **GAN integration strategy and 5-Fold data isolation:** The dataset is strictly partitioned using a 5-fold cross-validation strategy (divided into 60% Training, 20% Validation, and 20% Testing sets at each fold). The crux of this strategy is absolute adherence to the ''Zero Data Leakage'' principle. Specifically, the Testing set of each fold is kept fixed and completely isolated; absolutely no artificial images from the GAN network are allowed to leak into this set. This setup is a mandatory filter to ensure that all final evaluation metrics are objective and 100% accurately reflect the operational capability on candidates' actual answer sheets. For weight optimization, the Training set is configured under two comparative scenarios to precisely measure the efficacy of the generative adversarial network:
   - **Baseline scenario (Without GAN):** The models are trained on a baseline dataset where the minority *crossed-out* class is amplified to 1,500 samples exclusively using traditional, non-linear geometric augmentations.
   - **GAN-Integrated scenario (With GAN):** Exactly 1,000 synthetic *crossed-out* samples generated by the dynamically optimized DCGAN are directly incorporated into the training set, expanding the minority class to 2,500 samples per fold. This additional morphological diversity acts as a robust regularization mechanism, helping the model's decision boundary become more stable in regions with ambiguous features.
3) **Explicit Training Arguments and Loss Optimization:** To rigorously address class imbalance for models utilizing our custom PyTorch training loop (EfficientNet-B0 and ResNet50), optimization was performed using Adam (`lr=1e-4`, `batch=64`, `epochs=30`) with Early Stopping (`patience=10`). A dynamic weight tensor was calculated for the Cross-Entropy Loss function based on the exact per-fold class distribution:

$$W_i = \frac{N}{C \times N_i} \tag{6}$$

where N is the total number of training samples in the current fold, C is the total number of classes ($C = 3$), and N_i is the actual number of samples belonging to class i. This mathematical approach guarantees that

the neural network heavily penalizes misclassifications on the minority *crossed-out* class proportionally to its exact scarcity.

Conversely, to evaluate the efficacy of highly-optimized, "out-of-the-box" commercial frameworks, the YOLO26s-cls architecture was deployed using its native Ultralytics ecosystem. To guarantee full reproducibility, we explicitly document its configuration. Data feeding was managed via symlink-based dynamic routing to transition between the 1,500 and 2,500 minority-sample scenarios without duplicating storage. The model was initialized from the `yolo26s-cls.pt` checkpoint using the framework's default `seed=0` with `deterministic=True` and its standard distributed random sampler. The training arguments were aligned with the baseline constraints: `imgsz=128`, `epochs=30`, `batch=128`, `optimizer='Adam'`, `lr0=1e-4`, and `patience=10`. To match the photometric robustness of the PyTorch pipelines, we manually configured YOLOs HSV spatial augmentations (`hsv_s=0.3`, `hsv_v=0.3`, `hsv_h=0.01`). Regarding loss optimization, rather than employing custom dynamic class-weight injection, YOLO26s-cls evaluated the exact per-fold class counts (averaging 13,420 *Empty*, 6,588 *Confirmed*, and 1,500/2,500 *Crossed-out* samples) utilizing its default, native classification loss engine. This transparent setup allows a fair assessment between custom-engineered loss regularization and out-of-the-box framework optimizations.

V. EXPERIMENTS

A. IMPLEMENTATION DETAILS AND DATASET

1) TRAINING ENVIRONMENT

All neural network architecture experiments were conducted on the Google Colaboratory environment equipped with an NVIDIA Tesla T4 GPU hardware accelerator. This ensures fairness and consistency when measuring computational efficiency across models. The algorithms were developed using the open-source PyTorch framework and the Ultralytics library. Optimization utilized the Adam optimizer. Hyperparameters were strictly tailored to each architecture's requirements: batch sizes ranged from 32 to 128 depending on VRAM constraints, and base learning rates were set between 10^{-4} and 2×10^{-4}. The localization network was trained for 50 epochs, while classification networks were trained for a maximum of 30 epochs, incorporating an Early Stopping mechanism (patience = 10) to prevent overfitting. The experimental environment configuration is summarized in Table 1.

TABLE 1. Experimental environment configuration.

Component	Configuration Details	Notes
Deployment Platform	Google Colaboratory	Cloud environment
Hardware	NVIDIA Tesla T4 GPU Accelerator	16GB VRAM capacity
Library/Framework	PyTorch, Ultralytics	Open-source platform

2) DATA SOURCE

To ensure an objective and fair comparison with the baseline study, we inherited the OMR dataset published by Afifi & Hussain [5]. This dataset provides two specialized components serving different tasks within the system:

Full-sheet images dataset: Comprises 735 actual answer sheets totaling 815 pages (including both filled and blank sheets, spanning from the Exam 0 to Exam 5 structures), as shown in Fig. 4. To train the YOLO26s localization network, this dataset was partitioned at the page level into a 70% training set, 20% validation set, and 10% testing set.

Extracted ROIs dataset: To establish a uniform frame of reference when comparing the performance of the core classification models, we extracted individual image patches from the aforementioned 815 pages using the dataset's ground-truth bounding box coordinates. This extraction strategy guarantees that every ROI perfectly maps to its parent physical sheet ID, which is a mandatory prerequisite for our strict zero-leakage 5-fold cross-validation mechanism.

3) CLASSIFICATION DATASET DETAILS

The complete classification dataset comprises a total of 33,549 ROI images (with heterogeneous original sizes, which we standardized to 128 × 128 pixels via white-square padding). We chose this dataset because, unlike recent 2024-2025 studies that rely on private data, this is a publicly available benchmark that explicitly captures the severe class imbalance of real-world crossing-out behaviors. The dataset comprises three classes: *confirmed*, *crossed-out*, and *empty*. Sample images are illustrated in Fig. 5. The natural distribution of this dataset explicitly reveals the extreme data imbalance:

- **Confirmed class:** 10,980 images.
- **Empty class:** 22,367 images.
- **Crossed-out class:** Only 202 real images (accounting for roughly 0.6% of the total dataset).

To conduct training and evaluation for the classification models (EfficientNet-B0, YOLO26s-cls, ResNet50), we implemented a 5-Fold Cross-Validation protocol utilizing Sheet-level Stratified Sampling.

Testing set: To prevent data leakage and reflect the models' generalization capability with absolute accuracy, the data partitioning was enforced based on the physical IDs of the scanned sheets. In each fold, exactly 20% of the total sheets were completely isolated to form the Testing set. This ensures that no ROI extracted from the same physical sheet can simultaneously exist in both the training and testing sets.

Training and Validation sets: The remaining 80% of the sheets in each fold were further split at a 3:1 ratio to form the Training set (60% of total) and the Validation set (20%



FIGURE 4. Samples of answer sheets in the proposed dataset [5].

of total). To precisely measure the efficacy of the generative adversarial network, the training process was evaluated under two scenarios:

- **Baseline Scenario (Without GAN):** The training set for the minority *crossed-out* class (initially containing ∼120 real samples per fold) was amplified to exactly 1,500 images utilizing only traditional geometric and photometric augmentations.
- **DCGAN-Enhanced Scenario (With GAN):** 1,000 synthetic *crossed-out* images generated by the DCGAN were explicitly injected into the Training set of each fold (as shown in Fig. 6), expanding the *crossed-out* training samples to 2,500 images.

Crucially, the Testing and Validation sets in every fold remained in their original natural distribution and were completely devoid of any artificial GAN data.

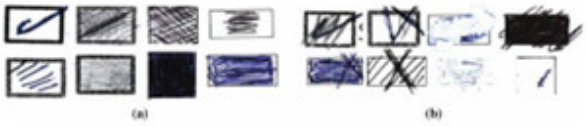


FIGURE 5. Samples of (a) confirmed and (b) crossed-out answers in the proposed dataset.

B. EVALUATION METRICS

To comprehensively evaluate the performance of the proposed system, we conducted measurements across two dimensions: the independent performance of the core classification model (on the ROI testing set) and the grading reliability of the entire system (end-to-end pipeline).

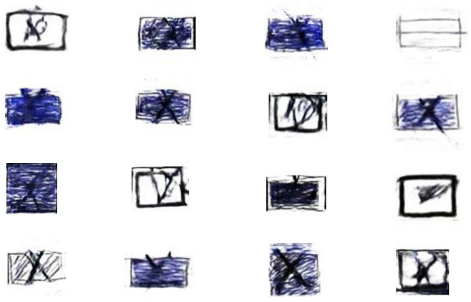


FIGURE 6. Samples of morphologically diverse crossed-out answers generated by the DCGAN.

1) CLASSIFICATION MODEL METRICS

The state classification performance at the individual bubble level was evaluated using a Confusion Matrix. For the multi-class classification problem (*Confirmed*, *Crossed-out*, *Empty*), the performance is rigorously quantified. The four primary metrics utilized include:

Overall Accuracy: Since the task evaluates three mutually exclusive states, the overall accuracy is calculated using the multi-class confusion matrix M. It is defined as the sum of the correctly predicted instances (the diagonal of the matrix)

divided by the total number of instances:

$$Accuracy = \frac{\sum_{i=1}^{3} M_{i,i}}{\sum_{i=1}^{3}\sum_{j=1}^{3} M_{i,j}} \tag{7}$$

Precision: The ratio of correct positive predictions to the total predicted positives for a specific class. This metric assesses the model's ability to avoid false recognitions.

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

Recall: The ratio of correct positive predictions to the actual total positives for that class. This metric is especially crucial for determining whether the model misses any *crossed-out* bubbles.

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

F1-Score: The harmonic mean of Precision and Recall, providing an objective evaluation of the model's performance on imbalanced datasets.

$$F1\text{-}score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

(Note: Due to the extreme disparity in sample sizes across classes, relying on the Weighted Average method would severely mask the model's true diagnostic behavior on the minority class. Therefore, we explicitly employed the ***Macro-Average*** *method for the overall Precision, Recall, and F1-Score metrics to ensure that the performance on the minority crossed-out class is equally weighted and transparently reported).*

Here, the True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN) parameters are derived locally from the Confusion Matrix when evaluated for a specific individual class (e.g., the *crossed-out* class) in a One-vs-Rest manner:

- **TP (True Positive):** The number of answer bubbles correctly predicted by the model as belonging to the class under consideration.
- **FP (False Positive):** The number of actual answer bubbles not belonging to the class under consideration, but incorrectly predicted into this class by the system.
- **FN (False Negative):** The number of actual answer bubbles belonging to the class under consideration but missed by the system and incorrectly predicted as another class.
- **TN (True Negative):** The number of answer bubbles correctly predicted by the system as not belonging to the class under consideration.

2) SYSTEM GRADING ACCURACY

Beyond evaluating the model on discrete cropped images, the most practical metric for an OMR system is its ability to output an accurate final score. Adhering to the evaluation standards of the baseline study, we applied two criteria:

Question-based Accuracy:

A multiple-choice question typically consists of multiple options (e.g., 4 bubbles A, B, C, D). The system is considered to have graded a question correctly if and only if it accurately classifies the states of *all* the constituent bubbles of that question. Accuracy is calculated according to the original paper [5]:

$$Accuracy_{question} = \frac{\text{number of correctly graded questions}}{\text{total number of questions}} \tag{11}$$

Sheet-based Accuracy:

In the context of high-stakes grading in educational systems, a system correctly grading 99 questions but missing 1 on the same test still results in a skewed final score for the candidate. Therefore, an answer sheet is considered "correctly graded" if and only if all questions on that sheet are recognized by the system without a single error. This accuracy is calculated as:

$$Accuracy_{sheet} = \frac{\text{number of correctly graded sheets}}{\text{total number of answer sheets}} \tag{12}$$

The $Accuracy_{sheet}$ metric serves as the clearest and most vital proof of the proposed system's commercial reliability, as it directly reflects the proportion of exams requiring absolutely no manual verification intervention.

C. RESULTS

1) CLASSIFICATION PERFORMANCE

To evaluate the classification performance for individual answer bubbles, we compared three architectures, namely ResNet50, YOLO26s-cls, and EfficientNet-B0, under two training scenarios: (i) the baseline model trained on the original dataset (Without GAN), and (ii) the model trained on an augmented dataset with an additional 1,000 DCGAN-generated images in each fold (With GAN). All results are reported as the mean and standard deviation over 5-fold cross-validation.

TABLE 2. Overall classification performance comparison of ROI classification models.

Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1-Score
YOLO26s-cls	Without GAN	99.42 ± 0.18	87.83 ± 5.45	85.70 ± 1.33	86.28 ± 2.67
YOLO26s-cls	With GAN	99.48 ± 0.16	88.05 ± 4.81	87.97 ± 2.02	87.88 ± 3.33
ResNet50	Without GAN	97.30 ± 0.70	71.93 ± 2.08	87.44 ± 1.14	74.36 ± 2.77
ResNet50	With GAN	97.46 ± 0.28	71.87 ± 1.03	87.63 ± 2.41	74.45 ± 1.42
EfficientNet-B0	Without GAN	99.47 ± 0.23	88.71 ± 5.33	89.31 ± 1.82	88.63 ± 3.19
EfficientNet-B0	With GAN	$\mathbf{99.54 \pm 0.16}$	$\mathbf{90.91 \pm 4.14}$	89.21 ± 3.09	$\mathbf{89.79 \pm 2.77}$

Table 2 shows that EfficientNet-B0 achieves the highest performance across most evaluation metrics. When trained with the DCGAN-augmented dataset, the model achieves an Accuracy of 99.54% ± 0.16%, Macro Precision of 90.91% ± 4.14%, Macro Recall of 89.21% ± 3.09%, and a Macro F1-Score of 89.79%±2.77%, outperforming both YOLO26s-cls and ResNet50. In comparison, YOLO26s-cls achieves an F1-Score of 87.88%, while ResNet50 achieves only 74.45%.

Although all three models achieve an Accuracy above 97%, this metric alone does not fully reflect the classification performance because the dataset is severely imbalanced among the three classes. The *empty* and *confirmed* classes account for the majority of samples, whereas the *crossed-out* class represents only a small proportion of the dataset. Therefore, Precision, Recall, and particularly the Macro F1-Score are considered more appropriate evaluation metrics for this task.

The incorporation of DCGAN-generated data consistently improves the performance of EfficientNet-B0 and YOLO26s-cls. Specifically, the F1-Score of YOLO26s-cls increases from 86.28% to 87.88%, corresponding to an absolute improvement of 1.60%, while EfficientNet-B0 improves from 88.63% to 89.79%, an improvement of 1.16%. In contrast, ResNet50 shows only a marginal improvement of 0.09% in F1-Score. This indicates that the ability to utilize DCGAN-generated samples depends considerably on the inherent feature extraction capability of the model architecture.

To directly evaluate the effectiveness of the proposed DCGAN-based data augmentation strategy, we further analyze the classification performance solely on the *crossed-out* class, which is the minority class augmented with synthetic samples during training. The Precision, Recall, and F1-Score for this class are reported in Table 3.

TABLE 3. Performance metrics on the minority crossed-out class.

Model	Dataset	Precision	Recall	F1-Score
YOLO26s-cls	Without GAN	0.6094	0.5792	0.5939
YOLO26s-cls	With GAN	0.6471	0.6535	0.6502
ResNet50	Without GAN	0.1605	0.6832	0.2599
ResNet50	With GAN	0.1667	0.6832	0.2680
EfficientNet-B0	Without GAN	0.6406	0.6881	0.6635
EfficientNet-B0	With GAN	**0.7340**	0.6832	**0.7077**

The results show that EfficientNet-B0 continues to achieve the best performance. The Precision of the *crossed-out* class increases from 0.6406 to 0.7340, while the Recall remains nearly unchanged (0.6881 and 0.6832). As a result, the F1-Score increases from 0.6635 to 0.7077, corresponding to a relative improvement of approximately 6.7%. For YOLO26s-cls, Precision increases from 0.6094 to 0.6471 and Recall increases from 0.5792 to 0.6535, resulting in an F1-Score boost from 0.5939 to 0.6502. In contrast, ResNet50 shows only a marginal improvement.

Figure 7 presents the overall confusion matrices of the three models before and after incorporating DCGAN-generated data. Most classification errors occur between the *confirmed* and *crossed-out* classes because both classes contain pencil marks inside the answer boxes and share similar visual characteristics. In contrast, the *empty* class is classified almost perfectly.

For ResNet50, the number of *confirmed* samples misclassified as *crossed-out* decreases from 476 to 420 after incorporating DCGAN-generated data. However, the number of *empty* samples misclassified as *crossed-out* increases from 246 to 270. Consequently, the Precision of the *crossed-out* class remains nearly unchanged, explaining the negligible improvement in the F1-Score of ResNet50.

YOLO26s-cls shows a more noticeable improvement. The number of correctly classified *crossed-out* samples increases from 117 to 132, while the number of *crossed-out* samples misclassified as *empty* decreases substantially from 29 to 13.

EfficientNet-B0 continues to achieve the best performance. After incorporating DCGAN-generated data, the number of *confirmed* samples misclassified as *crossed-out* decreases significantly from 59 to 35, corresponding to a reduction of approximately 40.7%. The number of *empty* samples misclassified as *crossed-out* also decreases from 19 to 15. Although the number of correctly classified *crossed-out* samples remains nearly unchanged (139 versus 138), the substantial reduction in false positive predictions pushes the Precision of the *crossed-out* class from 64.06% to 73.40%. These results indicate that the synthetic data not only helps the model recognize more minority samples but also enables it to learn a clearer decision boundary between visually ambiguous classes.

TABLE 4. Impact of DCGAN-generated data on the overall Macro F1-Score.

Model	Without GAN	With GAN	Improvement
YOLO26s-cls	86.28%	87.88%	+1.60%
ResNet50	74.36%	74.45%	+0.09%
EfficientNet-B0	88.63%	**89.79%**	+1.16%

Table 4 summarizes the improvement in the overall Macro F1-Score after incorporating DCGAN-generated data. Although the overall F1-Score improvement appears relatively small, the isolated results on the *crossed-out* class and the confusion matrices explicitly demonstrate that DCGAN effectively reduces class bias and enables the models to learn more robust decision boundaries. Ultimately, EfficientNet-B0 is confirmed as the optimal core component for the subsequent end-to-end evaluations.

2) END-TO-END SYSTEM GRADING ACCURACY

To transparently evaluate system performance and isolate error sources, we executed an automated grading process across all 735 actual answer sheets through two experimental scenarios: the Ideal Detection Pipeline scenario (using ground-truth coordinates) and the End-to-End Automated Pipeline scenario (using YOLO26s and ORB). Because our system architecturally decouples spatial detection from the per-sheet extraction path, this two-scenario evaluation follows naturally from the design, effectively isolating the variance introduced specifically by the geometric alignment process.

Scenario A: Ideal detection pipeline

In this scenario, to completely eliminate cumulative errors from automated localization and distortions from geometric alignment, we directly utilized the ground-truth bounding

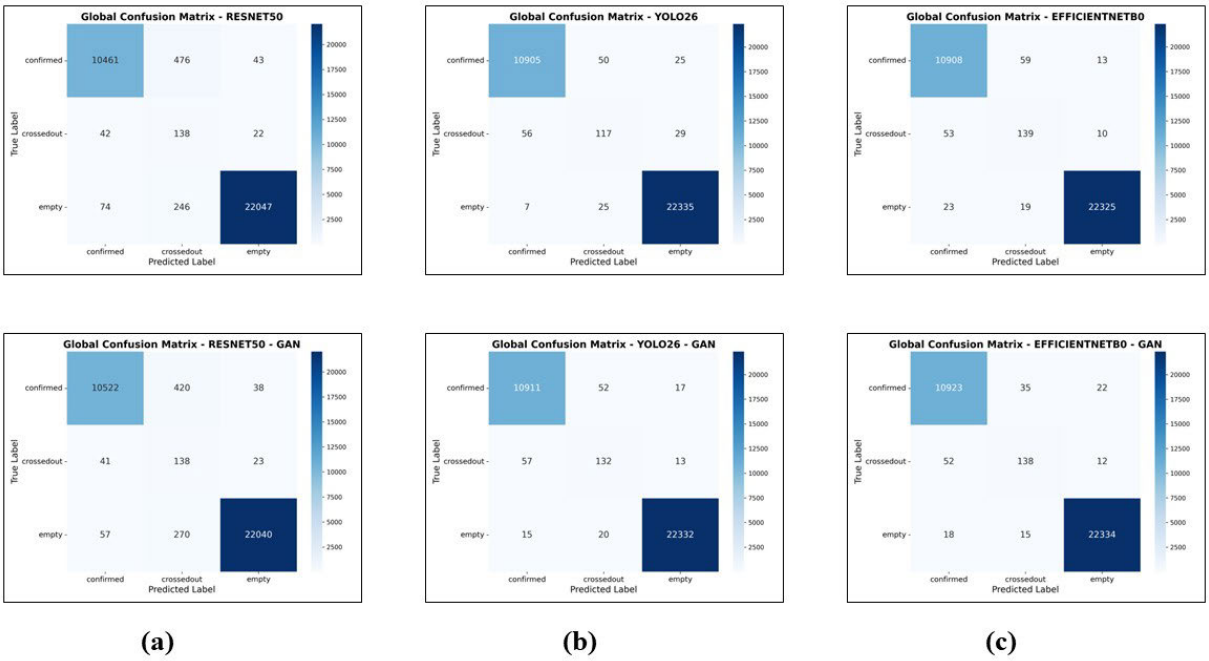


FIGURE 7. Global confusion matrices of ResNet50 (a), YOLO26s-cls (b), and EfficientNet-B0 (c) under the Without GAN (top row) and With GAN (bottom row) training scenarios.

boxes provided by the original authors to crop the multiple-choice bubbles. This method establishes an ''upper bound'' on the pure performance of the core classification models.

TABLE 5. System accuracy using ideal crop coordinates.

System Configuration	Q_Acc (%) (Mean ± Std)	S_Acc (%) (Mean ± Std)
CNN AlexNet (External Reference) [5]	99.78	97.69
YOLO26s-cls (without GAN)	98.45 ± 0.57	87.76 ± 1.60
YOLO26s-cls (with GAN)	98.48 ± 0.57	89.25 ± 2.23
EfficientNet-B0 (without GAN)	98.68 ± 0.75	88.71 ± 3.81
Proposed: EfficientNet-B0 (with GAN)	$\mathbf{98.82 \pm 0.67}$	$\mathbf{89.39 \pm 3.59}$

It is important to note that the external Afifi reference row in Table 5 was evaluated under their original protocol and data splits. While both our Scenario A and the Afifi baseline utilize pre-extracted ROIs, their evaluation setup differs from our strict 5-fold cross-validation with sheet-level isolation. Due to these methodological differences, this external number is presented strictly as a reference point rather than a direct algorithmic comparison. Our core contribution lies in removing the manual coordinate setup step while keeping accuracy in a highly usable range.

The experimental results from Table 5, evaluated under a 5-fold cross-validation protocol, provide crucial insights. A significant gap between Question Accuracy (Q_Acc) and Sheet Accuracy (S_Acc) can be observed across all configurations. For instance, while EfficientNet-B0 (with GAN) achieves a Q_Acc of up to 98.82%, the S_Acc is 89.39%. This disparity illustrates the strict error propagation effect characteristic of OMR: since an answer sheet contains dozens of questions, misclassifying a single ambiguous *crossed-out* mark as a *confirmed* bubble renders the entire sheet incorrectly graded.

The integration of DCGAN-generated data yields consistent improvements in mitigating these sheet-level failures. Notably, for the YOLO26s-cls architecture, the synthetic data successfully fortified the decision boundaries, elevating the mean S_Acc from 87.76% ± 1.60% to 89.25% ± 2.23%. By providing a richer feature space for minority edge cases, the GAN enables the model to rescue entire sheets. Ultimately, the proposed EfficientNet-B0 architecture paired with the GAN-augmented dataset demonstrates the highest ceiling performance, proving itself as a highly reliable core classifier.

Scenario B: End-to-End automated pipeline

This is the practical operational scenario of the proposed system. Crop coordinates are not pre-supplied but must be automatically inferred by the system via the YOLO26s network, followed by perspective alignment using the ORB algorithm. The results are presented in Table 6.

TABLE 6. End-to-end automated system accuracy (YOLO26s + ORB).

Model	Data	Q_Acc (%) (Mean ± Std)	S_Acc (%) (Mean ± Std)
YOLO26s-cls	Without GAN	98.36 ± 0.52	87.35 ± 2.07
YOLO26s-cls	With GAN	98.28 ± 0.52	88.16 ± 2.24
EfficientNet-B0	Without GAN	98.70 ± 0.71	88.84 ± 3.93
EfficientNet-B0	With GAN	$\mathbf{98.70 \pm 0.68}$	$\mathbf{88.98 \pm 3.92}$

To robustly evaluate the system's end-to-end reliability, a Wilcoxon signed-rank test was conducted on the paired Sheet Accuracy metrics across the 5 folds. The test yielded

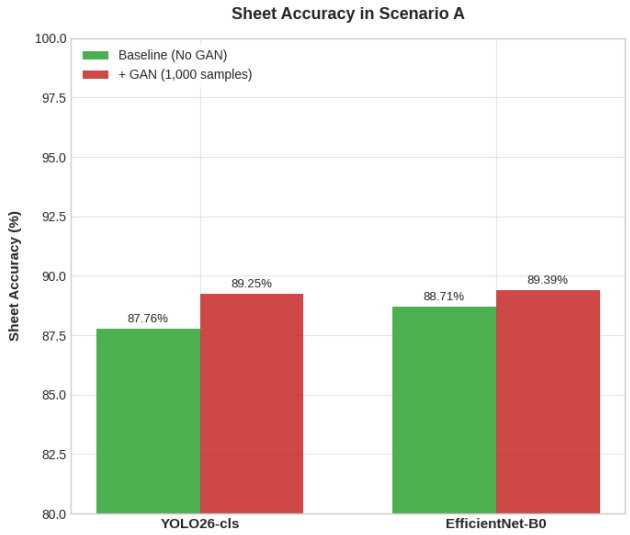


FIGURE 8. Sheet accuracy in Scenario A.

a p-value of 1.000, indicating that the marginal 0.14% improvement in the GAN-augmented EfficientNet-B0 configuration does not reach statistical significance. Therefore, while the generative augmentation demonstrably sharpens the minority-class decision boundary at the isolated ROI level, its overarching impact on the end-to-end sheet accuracy remains within standard run-to-run variance.

When transitioning to a practical operational pipeline without per-bubble template setup, the input images inevitably contain spatial noise, such as center shifts caused by automated cropping. Table 6 details the end-to-end performance under these conditions.

A comparative analysis reveals a remarkably marginal performance degradation. For the most optimal configuration, EfficientNet-B0 (with GAN), the mean Sheet Accuracy experiences a minimal decrease from 89.39% $\pm$ 3.59% (in Scenario A) to 88.98% $\pm$ 3.92% (in Scenario B). The slight decrease of approximately 0.41 percentage points between these scenarios validates the stability of the YOLO26s network, combined with the Container Suppression and Adaptive Grid algorithms, in accurately extracting the initial Master Coordinate Map. This provides a precise foundation for the subsequent geometric alignment and downstream classification on individual sheets. By analyzing the structural design of our pipeline, this minor discrepancy can be decomposed into two distinct factors:

- **Detection Contribution (Negligible):** Because the YOLO26s detector is executed exactly once on static blank templates to establish a constant Master Coordinate Map, its detection variance across actual student sheets is neutralized. Therefore, dynamic detection errors are architecturally eliminated from the inference pipeline.
- **Alignment Contribution (Primary Source):** Since detection errors are isolated by design, this 0.41-point degradation is predominantly attributable to the geometric flattening step (ORB Homography). When mathematically warping the perspective of varying student sheets to match the constant Master Coordinate Map, pixel interpolation artifacts occur. These micro-translations cause the fixed coordinates to crop the answer bubbles slightly off-center. This spatial noise slightly alters the morphological features of the strokes, causing the core classifier to misclassify a marginal fraction of highly ambiguous edge cases, further underscoring the critical need for generative data augmentation.

From a systems engineering perspective, trading off 0.41% accuracy to gain a fully automated pipeline without manual template setup is a highly favorable compromise.

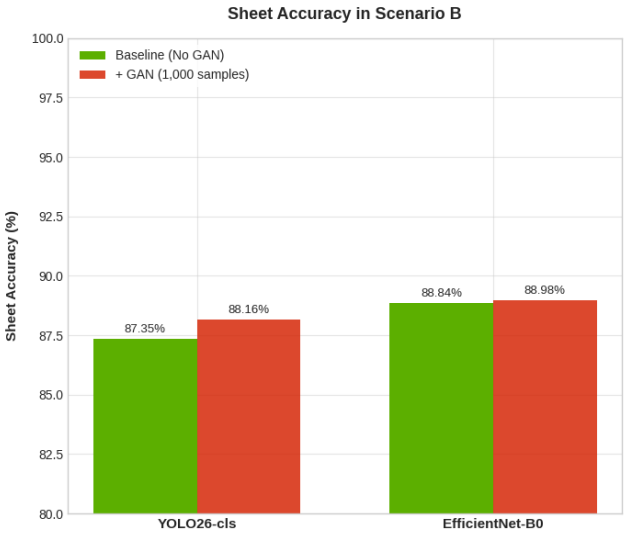


FIGURE 9. Sheet accuracy in Scenario B.

3) ON-DEVICE PROFILING AND END-TO-END LATENCY

To validate the practical feasibility of the lightweight mobile framework claim, it is essential to evaluate the system's end-to-end latency and memory footprint. We conducted an on-device profiling experiment utilizing a representative mid-range smartphone (Xiaomi Redmi Note 8 Pro, Android 11, 6GB RAM, MediaTek Helio G90T CPU) via the Android Studio Profiler.

Both the YOLO26s detection network and the EfficientNet-B0 core classifier were exported to the TensorFlow Lite (TFLite) format. To preserve maximum classification accuracy without the performance degradation typically associated with post-training quantization, the models were explicitly exported in the standard Float32 format. While the static model files are highly compact (YOLO26s occupies 36.4 MB and EfficientNet-B0 merely 15.4 MB), static file sizes do not fully reflect edge-deployment feasibility. Therefore, we monitored the application's RAM usage during a continuous 50-sheet grading loop at a standard high-resolution capture (1652 $\times$ 2338 pixels). The application

recorded a Peak Runtime Memory of ∼512.4 MB, which comfortably fits within the device's 6GB RAM allocation without triggering out-of-memory (OOM) exceptions.

We evaluated the total pipeline latency by decomposing it into specific stage-level metrics, utilizing 4 threads on the TFLite CPU delegate. The results are summarized in Table 7.

TABLE 7. End-to-end pipeline latency and on-device profiling.

Environment	Profiling Metric	Recorded Value
Cloud Baseline (Tesla T4 GPU)	Average Total Latency	0.77 s / sheet
	Peak Runtime Memory	∼512.4 MB
Mobile Edge (Redmi Note 8 Pro)	One-time Template Detection	∼2.20 s (Executed once)
	Warm-up Latency	∼3.01 s (First sheet only)
	Average Sheet Latency	2.95 ± 0.32 s
TFLite Configuration: 4 Threads, CPU Delegate	Stage 1: ORB Alignment	∼0.73 s
	Stage 2: ROI Extraction	∼0.01 s
	Stage 3: EfficientNet-B0	∼1.34 s

As detailed in Table 7, while the cloud-based GPU environment achieves an extremely fast processing speed of 0.77 seconds per sheet, the fully offline mobile deployment still maintains a highly efficient latency. Crucially, we explicitly separate the one-time Template Detection phase (which takes ∼2.20 seconds and executes only once per grading session) from the per-sheet processing loop. Inside the loop, the warm-up latency for the first sheet is slightly higher (∼3.01 seconds) due to TensorFlow Lite initialization.

For subsequent sheets, the average processing time stabilizes at 2.95 ± 0.32 seconds per sheet. This time is broken down into ORB Alignment (∼0.73 seconds), ROI Extraction and Preprocessing (∼0.01 seconds), and EfficientNet-B0 Inference (∼1.34 seconds), with the remainder accounting for local I/O operations (such as bitmap decoding) and scoring logic. For an end-to-end grading task, a processing time of under 3 seconds per image on an edge device thoroughly satisfies the real-time operational demands of educational applications.

4) ROBUSTNESS ON UNSEEN EXAM TEMPLATES

To evaluate the cross-institutional robustness of the proposed framework, we tested the system on a second, completely independent dataset consisting of 50 real-world exam sheets (Exam 6, illustrated in Fig. 10). This dataset utilizes a novel template format compared to the training dataset.

First, we evaluated the end-to-end automated grading pipeline under two image acquisition conditions to establish the system's fundamental cross-dataset generalization capability. To explicitly investigate the effect of generative augmentation on unseen templates, we report the metrics for both the baseline (Without GAN) and the GAN-augmented configurations. The results are detailed in Table 8.

The experimental results in Table 8 demonstrate the generalization capability of the baseline end-to-end pipeline. When evaluated on the entirely unseen Exam 6 template under standard scanning conditions, the baseline system

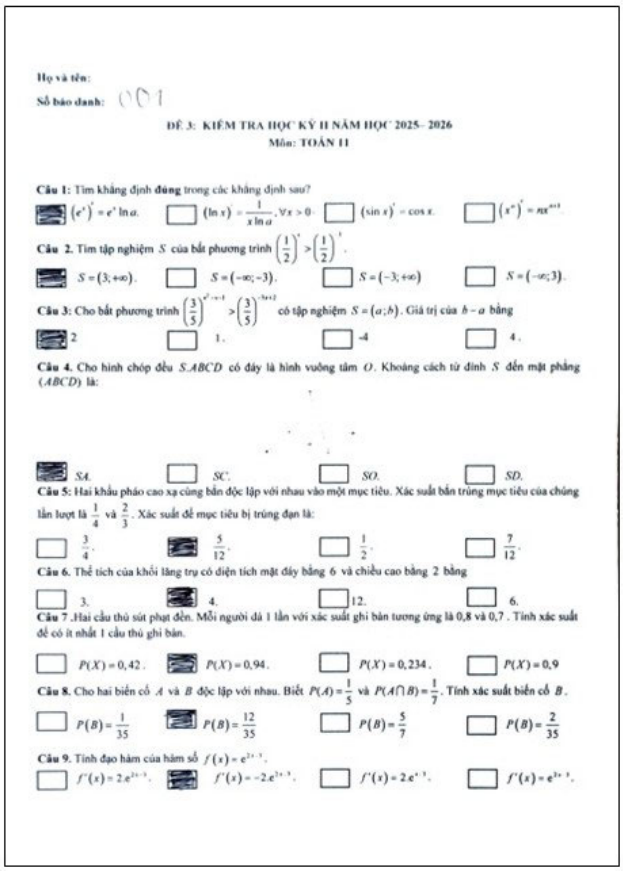


FIGURE 10. Samples of answer sheets in the unseen dataset (Exam 6).

TABLE 8. Cross-dataset evaluation on the unseen exam 6 template (Baseline vs. GAN-Augmented).

Image Source	Model Configuration	Q_Acc (%)	S_Acc (%)
Standard Scanner (App-assisted)	No GAN (Baseline)	94.49 ± 5.11	79.60 ± 7.13
	With GAN	75.96 ± 13.83	36.00 ± 15.75
Mobile Camera (Raw capture)	No GAN (Baseline)	77.69 ± 24.01	51.60 ± 28.96
	With GAN	74.09 ± 8.63	35.20 ± 16.16

maintained a robust Sheet Accuracy of $79.60\% \pm 7.13\%$. This confirms that the lightweight CNN architecture extracts generalizable optical features independently, without relying on fixed template layouts.

However, a significant performance degradation and higher variance were observed under mobile camera conditions, with the baseline Sheet Accuracy dropping to $51.60\% \pm 28.96\%$. This sensitivity highlights the challenges posed by uncontrolled environmental factors during mobile acquisition. It is important to clarify that the "Standard Scanner" results in Table 8 represent captures assisted by a mobile scanner application. This app-assisted approach effectively establishes our baseline for controlled conditions by automatically rectifying angles, equalizing illumination, and reducing blur. In contrast, the sharp drop to 51.60%

under "Mobile Camera" conditions represents raw, unguided captures using the native camera application, confirming that the system is currently vulnerable to "in-the-wild" variations. Severe perspective distortions from tilted capture angles, uneven illumination, and localized shadows impose significant strain on both the ORB-based geometric alignment and the subsequent feature extraction stages. Therefore, to deploy this pipeline reliably on mobile devices, the capture conditions should be controlled (e.g., via guided scanner applications) to ensure stable angles, even illumination, and minimal motion blur.

Crucially, incorporating the GAN-augmented configuration into this cross-template evaluation revealed an interesting phenomenon regarding generative data, which we term the "GAN Paradox". As explicitly shown in Table 8, the GAN-augmented model exhibited a substantial decrease in accuracy. Under standard scanner conditions, Sheet Accuracy declined from 79.60% to 36.00%, and under mobile conditions, it decreased to 35.20%. This contrast indicates that while DCGAN is exceptionally powerful at in-domain interpolation (mastering the specific pencil textures, lighting conditions, and bubble geometries of the original training dataset), it inadvertently forces the core neural network to overfit these localized physical traits. Consequently, the generative augmentation struggles with out-of-domain extrapolation when applied to entirely new templates. This finding suggests that for new cross-institutional deployments, relying on the robust baseline architecture initially yields more stable generalization than utilizing heavily regularized synthetic data.

5) COMPONENT ABLATION STUDY

To validate the structural safeguards of the spatial algorithms, an ablation study was conducted across the 815 pages. Disabling the Container Suppression algorithm resulted in 127 pages (15.6%) experiencing layout-extraction failures due to overlapping boxes. Re-enabling the module reduced layout failures to 3 pages. Furthermore, a sensitivity sweep of the adaptive grid threshold parameter (λ) from 0.4 to 0.8 revealed a stable error plateau between 0.5 and 0.8. While $\lambda \in [0.7, 0.8]$ yielded a reduction of a single row error, setting the threshold too high increases the risk of under-segmentation in densely packed templates. Therefore, $\lambda = 0.6$ was selected as a conservative safe margin, neutralizing intra-row alignment shifts without risking structural merging.

Regarding class imbalance, it is important to distinguish generative augmentation from standard loss adjustments. While techniques like Focal Loss scale gradients based on per-example difficulty, they do not inject novel visual intra-class variance. Because the bottleneck of the minority crossed-out class is the absence of complex erasure morphologies, DCGAN synthesizes these missing structural features in the pixel space, serving as a regularization mechanism.

VI. EVALUATION AND DISCUSSION

This section provides an in-depth discussion of the practical implications, architectural advancements, and the overall system design, as well as identifying the limitations of the proposed model in real-world application contexts.

A. AUTOMATION CAPABILITY AND ADAPTABILITY

The most significant contribution of the system at the architectural level is the complete elimination of manual per-bubble coordinate configuration, transitioning from rigid hard-coded grids to a dynamic template-learning mechanism, which overcomes a fundamental barrier in traditional OMR.

Bottleneck of traditional approaches: Traditional and semi-automated OMR methods (such as the baseline study by Afifi and Hussain [5]) operate based on static template matching mechanisms. This requires technicians to manually measure the physical distances of individual pixels to establish a rigid, hard-coded coordinate grid for each new exam format. This manual setup process is time-consuming and creates a significant bottleneck when attempting to deploy the system at scale.

In contrast, the proposed system applies the YOLO26s object detection network. Instead of searching for pixels at fixed positions, YOLO26s automatically interprets and localizes geometric entities (e.g., answer regions) based on their visual features. Combined with the Container Suppression algorithm and Adaptive Grid Sorting, the network can scan and automatically construct the entire Layout Map for the specified exam templates. This paradigm shift eliminates the need for per-bubble template setup. Educational institutions can use multiple flexible templates, while the system can automatically recognize and adapt to them efficiently, mitigating physical distortions such as skewed images or variations in capture distance.

B. CONTRIBUTION OF ADVERSARIAL DATA GENERATION

The performance gap between scenarios with and without GAN clearly demonstrates the role of DCGAN in shaping the feature space. Given that the crossed-out class accounts for a very small proportion in the natural data distribution, traditional data augmentation techniques (such as flipping, rotation, and noise injection) are insufficient. This is because these methods are essentially affine transformations applied to the existing pixel structure and cannot generate new structural characteristics of stroke patterns.

By generating 1,000 synthetic crossed-out samples, DCGAN addresses this limitation. The generated samples are not merely pixel perturbations but simulate the randomness and diversity in real human behavior, such as irregular strokes, partially erased marks, and complex overlapping markings. To quantitatively confirm this out-of-distribution generation capability, an automated evaluation against the unaugmented held-out testing sets was conducted, yielding a stable mean FID score of 204.86 ± 5.25 across the 5 folds. Coupled with the low SSIM values evaluated against the

original training data (proving the absence of memorization), this synthetic dataset provides rich intra-class diversity and acts as a strong regularization mechanism to reshape the decision boundary of the model.

Furthermore, while simpler imbalance-handling baselines could be considered, they fundamentally fall short in addressing the visual complexity of this computer vision task. Relying solely on weight-penalizing strategies—such as Focal Loss or our employed Weighted Cross-Entropy—only adjusts gradient scaling to heavily penalize minority class misclassifications. While effective mathematically, loss tweaking does not inject novel visual intra-class variance into the network, thus failing to overcome the physical absence of complex erasure patterns. Similarly, applying techniques like SMOTE (Synthetic Minority Over-sampling Technique) in the deep feature space is suboptimal for this specific domain. SMOTE creates new samples by linearly interpolating between existing minority feature vectors. However, the core challenge of recognizing crossed-out bubbles lies in capturing unpredictable visual morphologies, such as irregular strokes and complex overlapping ink. Linear interpolation in the feature space cannot synthesize these entirely new spatial and pixel-level geometric behaviors. Therefore, generative modeling via DCGAN is required because it operates directly in the pixel space, synthesizing novel and realistic erasure structures that act as a far superior regularization mechanism compared to feature-level mathematical interpolation or algorithmic loss adjustments.

In terms of classification performance, the effectiveness of this synthetic dataset is quantitatively demonstrated at the localized region level under the rigorous 5-fold cross-validation. For the proposed EfficientNet-B0 architecture, an analysis of the global confusion matrix reveals that the integration of GAN data effectively minimized model bias. Specifically, the number of false positives—where valid confirmed marks were incorrectly predicted as crossed-out—decreased notably from 59 to 35. This refinement of the decision boundary directly increased the Precision on the minority crossed-out class from 64.06% to 73.40%, elevating its specific F1-Score from 66.35% to 70.77%. Consequently, the overall Macro F1-Score of the core classifier also improved from 88.63% $\pm$ 3.19% to 89.79% $\pm$ 2.77%. However, transitioning this isolated enhancement into end-to-end grading reliability presents a more nuanced reality. In the fully automated operational pipeline (Scenario B), the GAN-augmented EfficientNet-B0 successfully recovered from ambiguous edge cases to achieve a Sheet Accuracy of 88.98%$\pm$3.92%. Yet, as confirmed by our rigorous statistical analysis, the overarching impact of this ROI-level gain on the final sheet-level accuracy remains within standard run-to-run variance, highlighting the strict error propagation effect inherent to OMR systems.

C. SYSTEM DESIGN TRADE-OFFS

The experimental results present two practical system design trade-offs regarding accuracy and deployment capability:

1) **Question-Level vs. Sheet-Level Accuracy Trade-off:** During the 5-fold evaluation in the fully automated pipeline (Scenario B), a notable statistical trend was recorded for the YOLO26s-cls architecture. Upon integrating GAN data, its Question-Level Accuracy (Q_Acc) slightly decreased by 0.08%, while the overall Sheet-Level Accuracy (S_Acc) concurrently increased by 0.81%. This trade-off is driven by the expansion of the decision boundary. The GAN data equips the network to recognize difficult minority edge cases, effectively rescuing entire exam sheets that would otherwise fail under the strict scoring logic. While this heightened sensitivity introduces a negligible fraction of isolated classification errors on individual bubbles, it noticeably elevates the overall sheet-level reliability. Notably, the specialized EfficientNet-B0 handles this boundary expansion more effectively, maintaining stable question-level metrics while stabilizing the peak sheet-level accuracy at 88.98% $\pm$ 3.92%.
2) **Cloud Inference vs. On-Device Mobile Latency:** While the cloud-based evaluation (NVIDIA Tesla T4 GPU) demonstrated an extremely fast average processing speed of 0.77 seconds per sheet, real-world deployment targets resource-constrained mobile devices. To validate this, both YOLO26s and EfficientNet-B0 were exported to the TensorFlow Lite (TFLite) format. To preserve maximum feature extraction accuracy, the models were kept in the unquantized Float32 format, resulting in a compact combined size of $\sim$52 MB. When deployed offline on a mid-range Android device (MediaTek Helio G90T CPU), the complete end-to-end processing time averaged 2.95 $\pm$ 0.32 seconds per sheet. Intentionally sacrificing cloud-level speed to run a 52 MB unquantized pipeline locally provides a feasible and secure system design for deployment in educational environments.

D. CURRENT LIMITATIONS OF THE SYSTEM

Although the proposed system performs well in automated settings, it exhibits certain limitations when dealing with complex real-world conditions and novel exam formats:

1) **Exact Failure Accounting in End-to-End Automation:** Correcting earlier metric artifacts, the final end-to-end evaluation of the 735 pooled sheets recorded an exact sheet failure rate of 11.02% (81 failed sheets). To systematically analyze the system's bottlenecks, these failures are distributed across three mutually exclusive categories:
   - *Geometric Alignment Errors (55 sheets):* Even with scanned document inputs, the ORB algorithm occasionally produced minor interpolation errors, resulting in slightly shifted bounding boxes. These misalignments introduced background noise or truncated bubble edges, which subsequently caused downstream classification failures.

- *Atypical Filling Misclassification (22 sheets):* The ROI extraction was perfectly aligned, but the EfficientNet-B0 model misclassified the state due to highly irregular human inputs, such as out-of-bounds cancellations, excessively faint pencil marks, or highly ambiguous marking styles.
- *Printing & Physical Artifacts (4 sheets):* The layout extraction failed entirely due to the physical condition of the sheets, such as wrinkled paper prior to scanning or severe printing ink smudges that obscured the structural markers.

2) **Cross-Institutional Generalization (The "GAN Paradox"):** To evaluate robustness, we tested the system on an independent dataset comprising 50 exam sheets with a novel template format. The baseline (Without GAN) EfficientNet-B0 model generalized reasonably well, maintaining a Sheet Accuracy of 79.60% ± 7.13% under standard scanning conditions. However, the GAN-augmented model exhibited a noticeably lower accuracy, dropping significantly to 36.00% ± 15.75%. This reveals a "GAN Paradox": while DCGAN effectively captures the specific pencil textures and paper backgrounds of the training set (in-domain), it inadvertently causes the core classifier to overfit these localized physical traits. Consequently, the generative augmentation struggles with out-of-domain extrapolation on new templates. For cross-institutional deployments, relying on the robust baseline architecture currently yields safer generalization.
3) **Uncontrolled Mobile Capture and Alignment Bottleneck:** As detailed in Table 8, there is a severe performance degradation when transitioning from controlled scanning to unguided mobile captures on unseen templates. For the baseline model, sheet accuracy drops from 79.60% ± 7.13% to 51.60% ± 28.96%. This vulnerability is further compounded in the GAN-augmented configuration, where the accuracy collapses from 36.00% ± 15.75% down to 35.20% ± 16.16% under mobile conditions. This substantial gap points squarely at the ORB geometric alignment step as the primary bottleneck when dealing with severe "in-the-wild" perspective and illumination distortions. When the spatial alignment is slightly off, the resulting cropping noise severely disrupts the highly-regularized decision boundary of the GAN-trained classifier. Rather than a mere caveat, addressing this gap provides a clearly identified next step for our research. Upgrading the spatial synchronization module is our primary focus for future work to stabilize unconstrained mobile deployments.

VII. CONCLUSION

This study has proposed and implemented a fully automated, end-to-end Optical Mark Recognition framework. By integrating the YOLO26s object detection architecture with the Box-in-Box noise filtering algorithm and Adaptive Grid Sorting, the system eliminates the bottleneck of traditional approaches that require per-bubble template setup, thereby enabling scalability across various exam formats.

Furthermore, this study addresses the extreme imbalance in the natural data distribution of the *crossed-out* class. By applying DCGAN to generate synthetic data, combined with the feature extraction capability of the EfficientNet-B0 core architecture, the system successfully sharpens the minority-class decision boundary. Evaluated under a rigorous 5-fold cross-validation protocol in a fully automated pipeline, the proposed model achieves a Question Accuracy of 98.70% ± 0.68% and a strict Sheet Accuracy of 88.98% ± 3.92%. While the generative augmentation consistently improves localized ROI-level precision, the statistical analysis confirms that the overarching end-to-end sheet-level gains remain within run-to-run variance. Crucially, on-device profiling confirmed that the entire unquantized framework (∼52 MB) executes completely offline on a mid-range smartphone CPU in an average of 2.95 ± 0.32 seconds per sheet, demonstrating a practical balance between real-time performance and high-stakes reliability.

Although the current system achieves strong performance on static image inputs, future work can be extended through four main directions:

- **Edge AI Optimization:** Applying advanced model quantization techniques (INT8/FP16) to further optimize mobile inference latency while maintaining precision.
- **Domain Adaptation:** Investigating domain-adaptation strategies to overcome the "GAN Paradox" and improve the cross-institutional generalization of generative augmentations on unseen templates.
- **Geometric Alignment Enhancement:** Upgrading the traditional ORB-based alignment module with advanced deep learning-based feature matching networks (such as LoFTR, or SuperPoint and SuperGlue) and Deep Homography Estimation. These detector-free and attention-based mechanisms will robustly handle severe blur and perspective distortions, effectively closing the performance gap between controlled scanning and unguided mobile captures.
- **Multimodal Recognition Capability:** Integrating Handwritten Text Recognition (HTR) models to automatically extract unstructured information such as candidate names, exam codes, or signatures, instead of being limited to bubble-based analysis.

All source code and experimental datasets have been made publicly available on GitHub [37], fostering reproducibility and facilitating further advancements in this area.

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